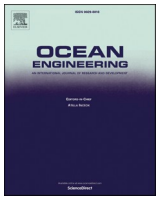
Ocean Engineering 294 (2024) 116766

Contents lists available at ScienceDirect

Ocean Engineering

journal homepage: www.elsevier.com/locate/oceaneng



Research paper

Collision evasive action timing for MASS using CNN–LSTM-based ship trajectory prediction in restricted area

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ARTICLE INFO

Keywords:
Ship collision
Collision avoidance
COLREGS
Maritime autonomous surface ship
CNN–LSTM
Artificial intelligence

ABSTRACT

One of the most important factors for the development of maritime autonomous surface ships (MASSs) is collision avoidance. Various artificial intelligence models have been applied to collision avoidance; however, they have a limited ability to understand the International Regulations for Preventing Collisions at Sea (COLREGS), which are qualitative in nature and do not provide specific timings for ships to take action. Herein, we quantified Rules 8, 16, and 17 of the COLREGS to help MASSs understand them. Further, we proposed specific timings for the engine control of a give-way ship to avoid imminent collision and the collision avoidance cooperative actions of a stand-on ship based on the sea conditions in the Singapore Main Strait. To evaluate the proposed approach, we used a convolutional neural network–long short-term memory (CNN–LSTM) model to predict ship trajectories in different scenarios and performed collision assessment according to the computed distance at collision. Two types of own ships were assessed: one with a good course changing performance and one with a poor curse changing performance.

1. Introduction

Advances in artificial intelligence (AI) can lead to the development of maritime autonomous surface ships (MASSs) (Pietrzykowski et al., 2022). The International Maritime Organization is preparing regulations for MASS operation, and many researchers are investigating MASS-related AI technology (Chun et al., 2021). Japan announced that it successfully conducted a field experiment of MASS operation (MEGURI, 2022; Negenborn et al., 2023), and Norway conducted a field experiment with a 120-TEU autonomous ship (Smartmaritime, 2020; Yara, 2020; Negenborn et al., 2023).

The critical factor for realizing practical MASS application is collision avoidance (Gao and Shi, 2020). Many researchers have applied various AI models to develop collision avoidance models (Cheng and Zhang, 2018; Wang et al., 2019, 2020; Liu et al., 2019; Li and Zheng, 2021; Yan et al., 2021). Cheng and Zhang (2018) presented an obstacle avoidance model that uses the concise deep reinforcement learning obstacle avoidance algorithm with a deep Q-networks architecture. Ning et al. (2020) presented a collision avoidance model that uses a genetic algorithm to apply the evolutionary principles. Fiskin et al. (2021) presented a collision avoidance model that combines a genetic algorithm with fuzzy logic. Wang et al. (2020) fused a hidden obstacle avoidance logic layer with an observable and predictable control layer to construct a collision avoidance decision process that improves the obstacle avoidance success rate in multi-ship encounter scenarios. Guo et al. (2020) presented a collision avoidance model based on the chaotic and sharing-learning particle swarm optimization algorithm, which is a biological heuristic algorithm that improves upon the particle swarm optimization (PSO) algorithm. Xu et al. (2020) presented a collision avoidance model based on the International Regulations for Preventing Collisions at Sea (COLREGS) and deep reinforcement learning (DRL). Woo and Kim (2020) studied collision avoidance using DRL and the semi-Markov decision process. Namgung and Kim (2021) presented a collision risk assessment model that uses an adaptive neuro-fuzzy inference system (ANFIS), which combines fuzzy logic with an artificial neural network. Lee and Park (2022) presented a dynamic ship domain model that combines a ship domain based on seafarers' awareness of safe distances (Lee et al., 2021) in a restricted area with ANFIS.

Many AI-based collision avoidance studies have used DRL to build autonomous navigation systems that can handle multi-ship collision avoidance to improve the autonomy and intelligence level of MASSs (Wang et al., 2019; Zhang et al., 2019; Chen and Chen, 2020). However, the above studies are limited in their ability to provide a specific timing for when to take actions based on Rules 16 and 17 of the COLREGS. At

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https://doi.org/10.1016/j.oceaneng.2024.116766
Received 29 August 2023; Received in revised form 18 December 2023; Accepted 13 January 2024
Available online 22 January 2024
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Symbols definitions

CDC	Computed distance at collision
CA	Collision assessment
OS	Own ship
TS	Target ship
TSS	Traffic separation scheme
VOS	Virtual OS
δ_{OS}	OS rudder angle
$t_{\psi_{os}=90^\circ}$	Time for OS heading to reach 90°
t_s	Simulation time
λ	Ship trajectory
λ_s	Ship trajectory obtained by simulation
λ_{cl}	Ship trajectory predicted by CNN–LSTM
λ_o	Ship trajectory not reflecting sea conditions
θ_{ot}	Intersection heading angle between the OS and TS
L	Ship length
AD	Actual distance between the OS and TS
θ_{ic}	Inevitable collision angle of θ_{ot}
β	Bearing angle between the OS and TS
x_{os}, y_{os}	Coordinates of the OS in an Earth-fixed coordinate system
x^f_{os}, y^f_{os}	Coordinates of the OS forecastle point in the Earth-fixed coordinate system
x_{vos}, y_{vos}	Coordinates of the VOS in the Earth-fixed coordinate system
x^f_{vos}, y^f_{vos}	Coordinates of the VOS forecastle point in the Earth-fixed coordinate system
x_{ts}, y_{ts}	Coordinates of the TS in the Earth-fixed coordinate system
ψ_{os}, ψ_{ts}	Headings of the OS and TS
x_c, y_c	Coordinates of the collision point in the Earth-fixed coordinate system
RP	Rendezvous point
d_{ca}	Difference between the CDC and AD
d_{df}	Difference between λ_s and λ_o
d_c	Distance at collision
t_c	Time of collision
τ_{action}	Timing for ship action

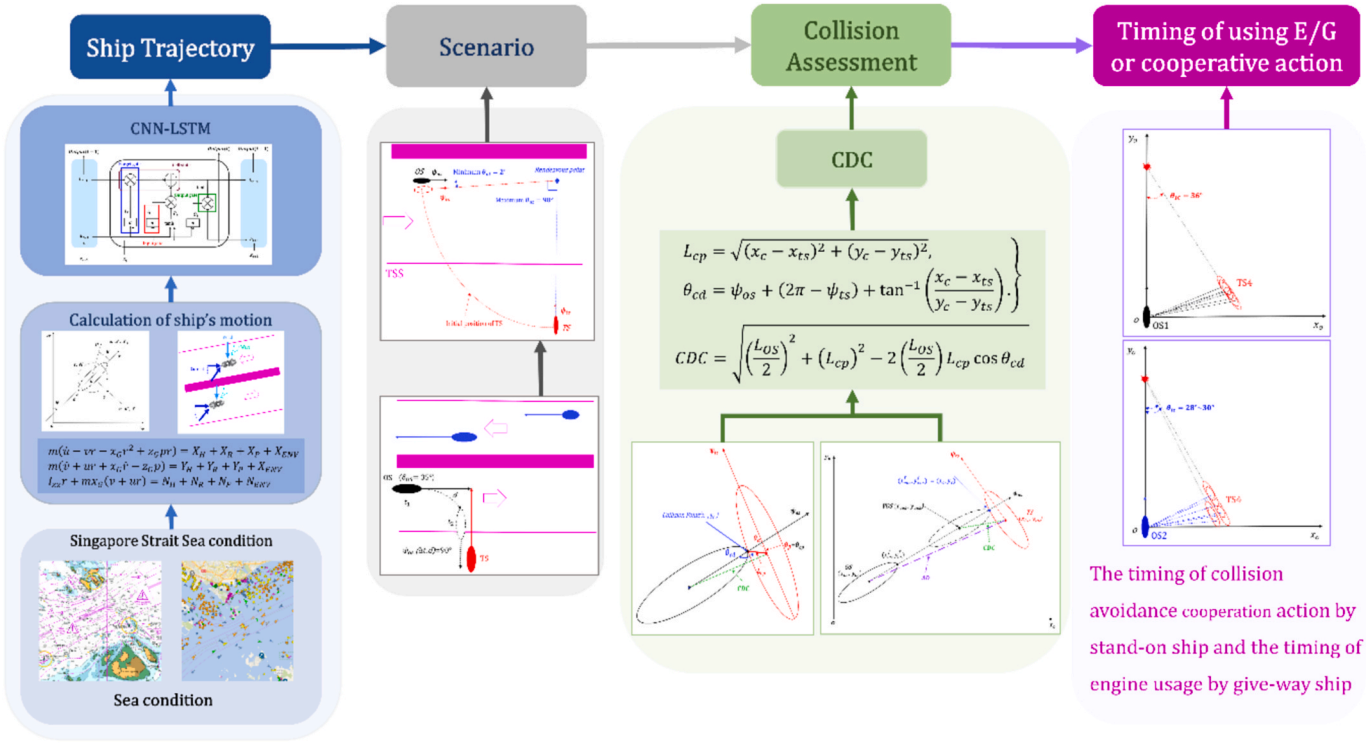


Fig. 1. Research flowchart.

sea, a ship can avoid collision in two ways: by changing course to avoid the target ship (TS) or by using its engine to adjust its speed. Thus, further research is needed on determining the specific timing for when to take an action (Porathe, 2019). Huang et al. (2020) described action lines as the last chance for an own ship (OS) to avoid collision. They suggest that an action line should be determined by simulations rather than expert judgment and that the TS should be assumed to maintain its initial speed while the OS takes evasive action. This assumption is based on simulation results on the critical distance (Szlapczynski et al., 2018; Krata and Montewka, 2015). Baldauf et al. (2017) also defined the action line as the last chance for evasion. However, these studies did not provide a specific timing for a ship to take an action.

Lee and Furukawa (2022) confirmed that there are situations where a collision cannot be avoided using the rudder of the give-way ship alone. However, they used ship trajectories that did not reflect sea conditions and subjectively set the OS and TS. According to COLREGS, in

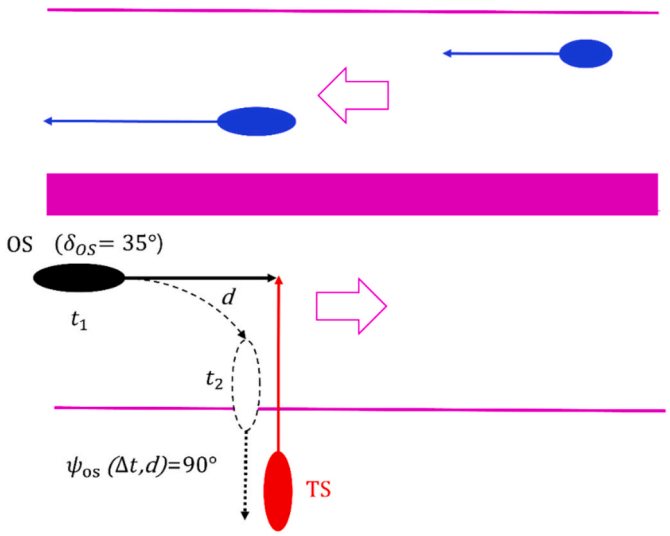


Fig. 2. Diagram for scenario setup.

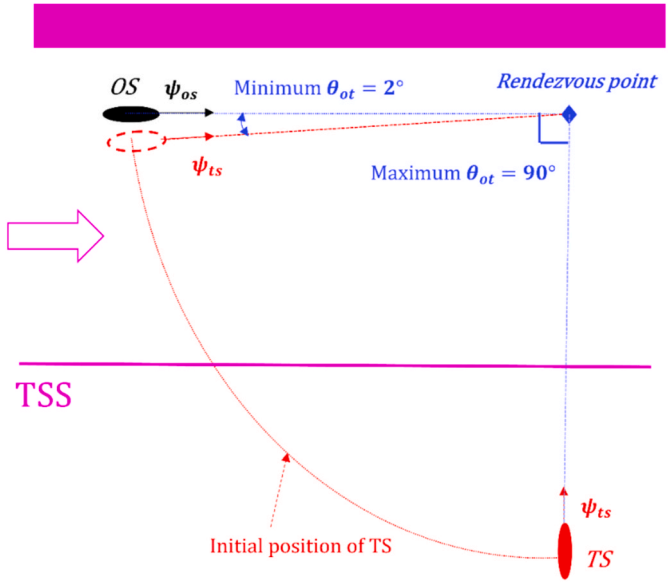


Fig. 3. Setting the initial position between ships.

Table 1
Parameters of the subject ships.

Ship parameter	OS1 (Bulk Carrier)	OS2 (Container Carrier)
Length, L (m)	233	274
Breadth, B (m)	32.30	32.30
Draft, d (m)	13.80	12.70
Block coefficient, C_b	0.88	0.65
Speed (knots)	15.75	25.00

their situation the give-way ship would use its engine and the stand-on ship would perform a cooperative action to avoid collision.

Of course, research has also been conducted on the collision avoidance action of ships based on AIS data (Rong et al., 2022). However, it did not provide specific numerical values for how collision avoidance action should be performed when two ships are very close, based on ship's maneuverability.

Herein, we developed an approach for determining the specific timing for collision avoidance actions through the engine control of the give-way ship and cooperative action by the stand-on ship considering the sea conditions in the ***Singapore Main Strait*** for different ship types. Determining a specific timing is important because of the vagueness of the COLREGS (Zhou et al., 2020; Wróbel et al., 2022), which do not provide the quantitative information on safe distances, safe speeds, or collision avoidance timing (He et al., 2017). Future MASSs will need to operate according to the COLREGS. However, while humans can understand the qualitative COLREGS and make appropriate judgments, current AI technology has difficulty and needs to be trained on quantitative data.

Herein, we calculated the ship trajectory of a model ship and predicted its trajectory using a convolutional neuron network–long shot term memory (CNN–LSTM) approach, which accurately processes time series data (Yan et al., 2021). the predicted ship's trajectory by CNN-LSTM, the collision assessment was performed by CDC (Computed Distance at Collision) in a very close situation between two ships.

The rest of paper is organized as follows. Section 2 describes the methodology of this study. How the collision scenario is constructed, how the ship's trajectory is calculated, and CNN-LSTM for trajectory prediction and CDC will be described. Section 3 will analyze the results of the collision assessment, and Section 4 will conclude and discuss the results of this study.

2. Methodology

Fig. 1 shows our proposed approach. The ship trajectory (λ) is obtained from the ship motion equations that consider the sea conditions. CNN–LSTM model is used to predict the obtained λ data. The scenario is configured by considering the traffic separation scheme (TSS), and the timing for actions is determined according to CDC. And in this study, the distance between the ship's trajectory and the sea chart is converted to meters.

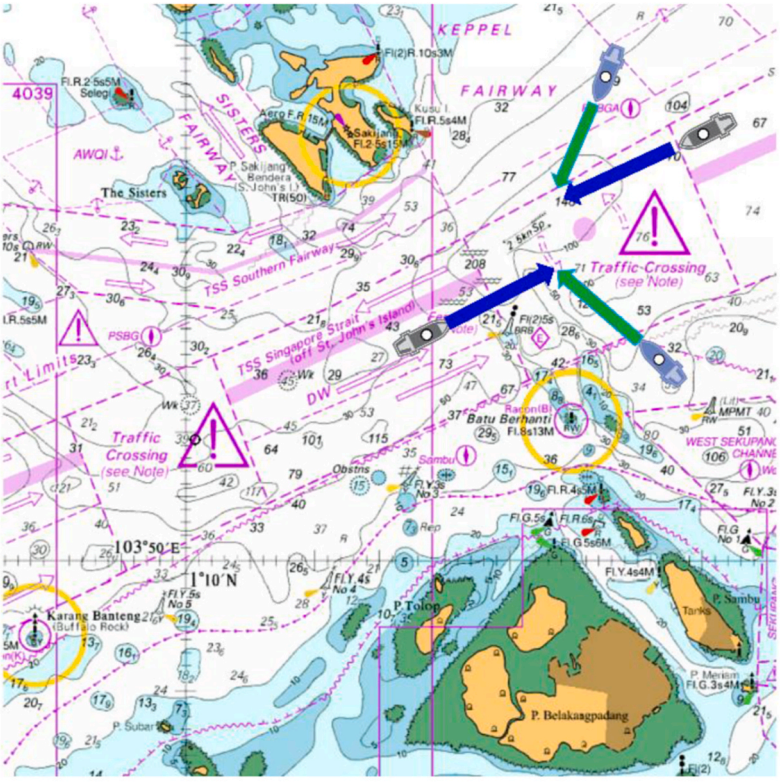
2.1. Scenario

Herein, we limited the scope to the TSS of the ***Singapore Main Strait***. We limited the scope to a specific area because the maritime environment is so complex that developing a universal model that is suitable for all sea areas is extremely difficult (Chai et al., 2017). This TSS was selected because it is difficult to navigate (Inoue, 2000; Chai et al., 2017) owing to the heavy traffic volume and narrow route (Kang et al., 2019).

Fig. 2 shows the situations that were considered to set up the scenario.

The black ellipse represents the OS, and the red ellipse represents the TS. δ_{os} denotes the rudder angle of the OS, ψ_{os} denotes the heading angle of the OS, t is the time the OS sailed, and d is the distance the OS sailed. According to Rule 10 of the COLREGS, the OS and TS are in a crossing situation. The OS is the give-way ship, so it should avoid the TS by turning to starboard. In the worst case, the OS may choose to change course by 90° and go off the TSS. Thus, the simulation time (t_s) was defined as the time it took for the OS to change its heading by 90° ($\Delta t_{\psi_{os}=90°}$).

Fig. 3 shows the initial positions of the OS and TS considering t_s. The black ellipse is the OS, the red ellipse is the TS, ψ_{os} and ψ_{ts} are the heading angles of the OS and TS, respectively, and θ_{ot} is the intersection heading angle between the OS and TS. The rendezvous point (RP)



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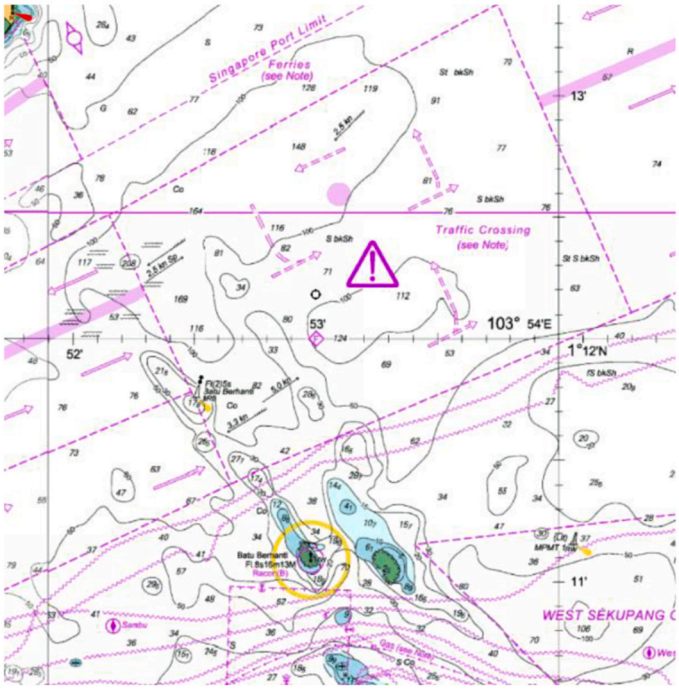

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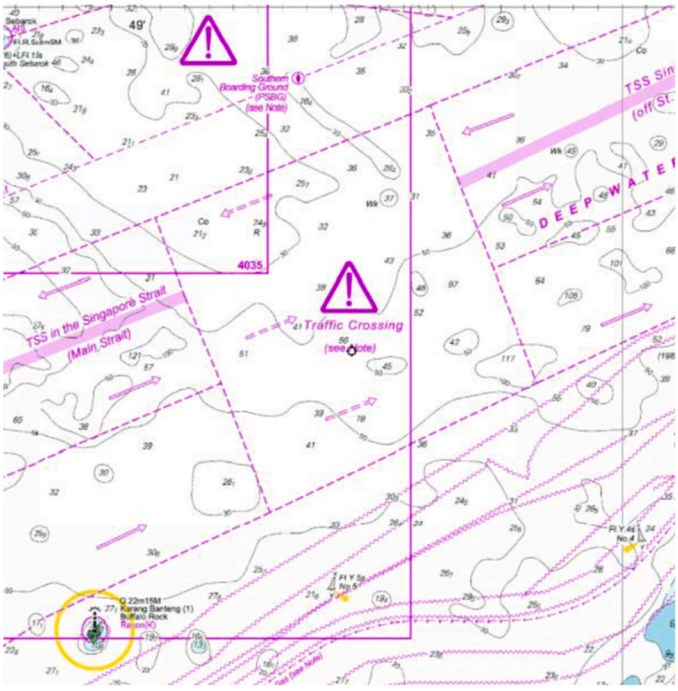


(c)

Fig. 4. Singapore Main Strait Sea chart.

Table 2
Sea conditions applied in the simulation.

	02:00, Feb 19, 2023	02:00, Feb 23, 2023	08:00, Feb 24, 2023
Wind direction	N	N	N
Wind speed (kt)	12	19	17
Wave direction	N	N	NNE
Wave (m)	1.4	1.7	3.7
Current direction	WSW	W	W
Current speed (kt)	1.8	0.8	0.8

Table 3
Scenario cases.

	Direction	ψ_{os}	OS1	OS2
Case 1	Eastbound	65 °	02:00, 19 Feb	02:00, 19 Feb
Case 2	Eastbound	65 °	02:00, 23 Feb	02:00, 23 Feb
Case 3	Eastbound	65 °	08:00, 24 Feb	08:00, 24 Feb
Case 4	Westbound	265 °	02:00, 19 Feb	02:00, 19 Feb
Case 5	Westbound	265 °	02:00, 23 Feb	02:00, 23 Feb
Case 6	Westbound	265 °	08:00, 24 Feb	08:00, 24 Feb

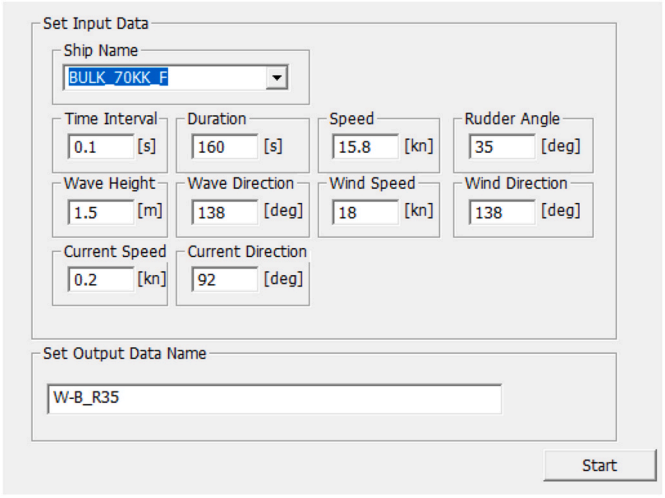


Fig. 5. Simulation program UI.

represents the point where the two ships collide after t_s if there are no disturbances such as wind, currents, and wave. The initial position of the TS considering the RP is indicated by the red curved dotted line. In simulations, we set the TS to sail toward the RP without changing speed and course. As the give-way ship, the OS must use its rudder to avoid the TS. According to the COLREGS, the evasive maneuver should be made toward the starboard side. If the OS cannot avoid collision with the TS by using the maximum rudder angle (35 °) to the starboard side, then it cannot act alone to avoid collision. Therefore, we only considered cases with an OS rudder angle of 35° in experiments. For ease of calculation, the two ships were represented as ellipses, where the coordinates of the ships were represented by the center points of the length and width of each ellipse. The OS was assumed to be a bulk carrier or container ship, which are the easiest or most difficult to change course according to the C_b coefficient (Ha and Gourlay, 2017; Lee et al., 2021).

Table 1 presents the details of OS1 and OS2. Four TSs with different lengths (L) and breadth (B) were selected based on monitoring data of ships sailing in the ***Singapore Main Strait*** (TS1: 30m/7m; TS2: 180m/30m; TS3: 280m/40m; TS4: 400m/60m). The speeds of the TS and OS were set to be the same.

2.2. Sea conditions

The ***Singapore Main Strait*** is a very complex area with ships sailing between Singapore and Malaysia entering the TSS, high-speed ferries heading to anchorages or Malaysia, and very large container ships and large tankers sailing between Europe and Asia that create various traffic situations.

Fig. 4 shows a sea chart of the ***Singapore Main Strait***. The TSS is in shallow waters, and the maximum width is < 1.3 nautical miles. There are separate routes for ships with the deepest draft, which can lead to very complex traffic. Moreover, as shown in Fig. 4 (a), it is a dangerous area where many crossing situations can happen because many ships can enter or cross the TSS. We obtained sea conditions from Applied Weather Technology Inc. (Sunnyvale, CA). The 6-h sea condition data were reviewed for two weeks in 2023: 18–24 February and 9–16 March. The worst conditions were then applied to our simulations. In summary, the wind in the ***Singapore Main Strait*** was mainly N to NNE with a minimum speed of 9 kt and a maximum speed of 19 kt. Waves were NNW to NE with a minimum height of 0.4 m and a maximum height of 3.7 m. The current was W to NE with a minimum speed of 0.2 kt and maximum speed of 1.8 kt. Table 2 presents the days with the strongest current, wind, and waves that were selected for the simulation.

The three selected days were each divided into two cases: eastbound and westbound ships sailing into the TSS. In the eastbound cases, ψ_{os} was set to 65°. In the westbound cases, ψ_{os} was set to 265°. Table 3 presents the scenario cases according to the TSS sailing direction and sea conditions.

2.3. Ship trajectory data

For CNN–LSTM to predict ship trajectories, it should be trained on real ship trajectories. However, obtaining real ship trajectory data was virtually impossible because ships rarely apply the maximum rudder angle of 35° unless in a critical situation. In addition, turning the ship rudder by 35° in a narrow area such as the ***Singapore Main Strait*** would likely turn into another dangerous situation. Therefore, we used a ship motion simulation program developed by Korea Research Institute of Ship and Ocean Engineering (KRISO) and STR (2020) to calculate the ship trajectory data. Fig. 5 shows the UI of the simulation program used in this study.

The simulated ship trajectories (λ_s) were obtained to reflect the sea conditions and factors affecting the rudder, propeller, and hull.

$$\left.\begin{aligned} m(\dot{u} - vr - x_G r^2 + z_G pr) &= X_H + X_R + X_P + X_{ENV} \\ m(\dot{v} + ur + x_G \dot{r} - z_G \dot{p}) &= Y_H + Y_R + Y_P + X_{ENV} \\ I_{zz}\dot{r} + mx_G(\dot{v} + ur) &= N_H + N_R + N_{ENV} \end{aligned}\right\}, \quad (1)$$

where.

m, m_x, m_y: hull mass, components of added mass of a ship in x- and y-axes
I_{zz}: moment and added moment of inertia of a ship around z-axis
u,: components of ship speed in x- and y-axes
p,r: Angular velocity around x- and z-axes.
G: center of gravity of ship
H, R, P and Env: hull, rudder, propeller and environment component such as wind, wave and current

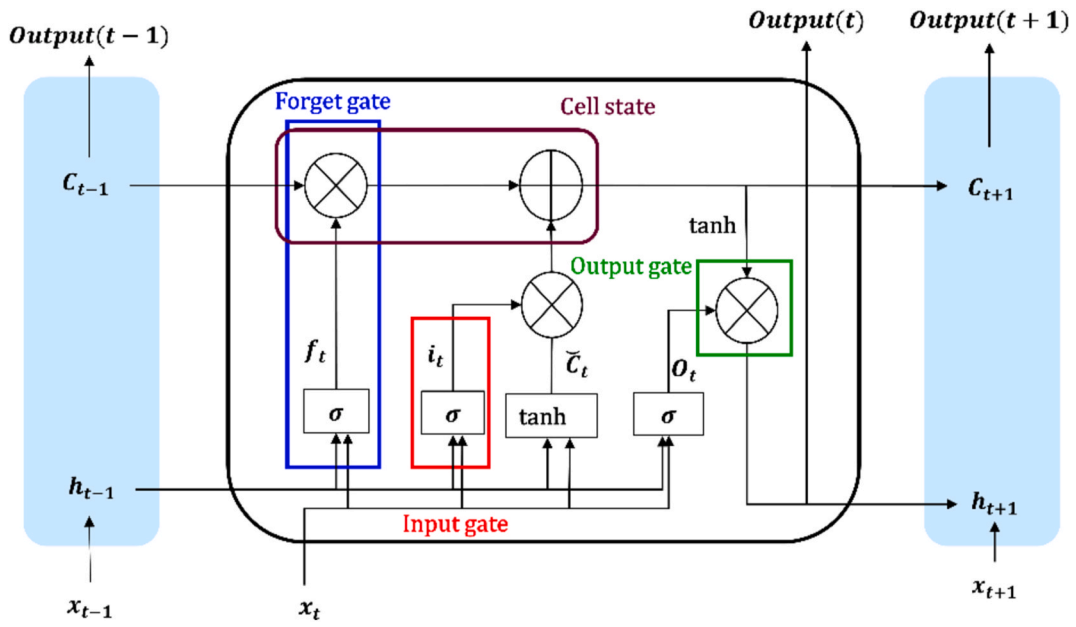


Fig. 6. Lstm schematic.

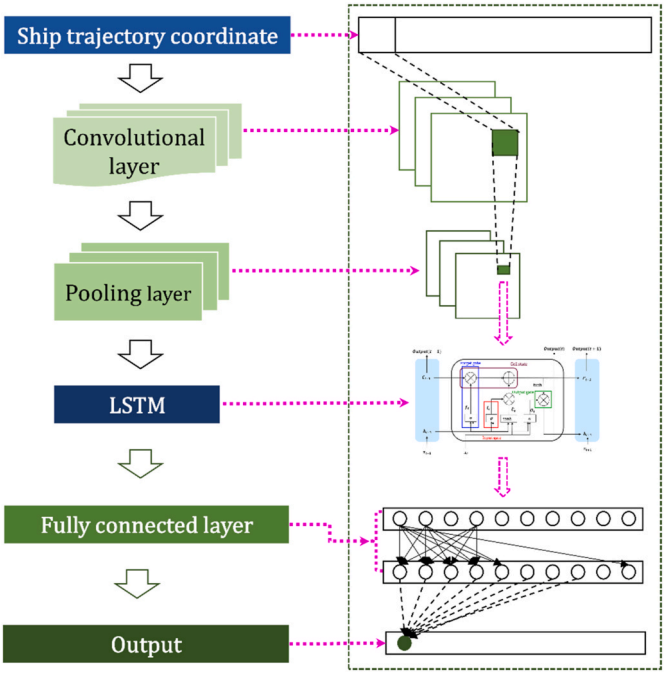


Fig. 7. CNN–LSTM architecture.

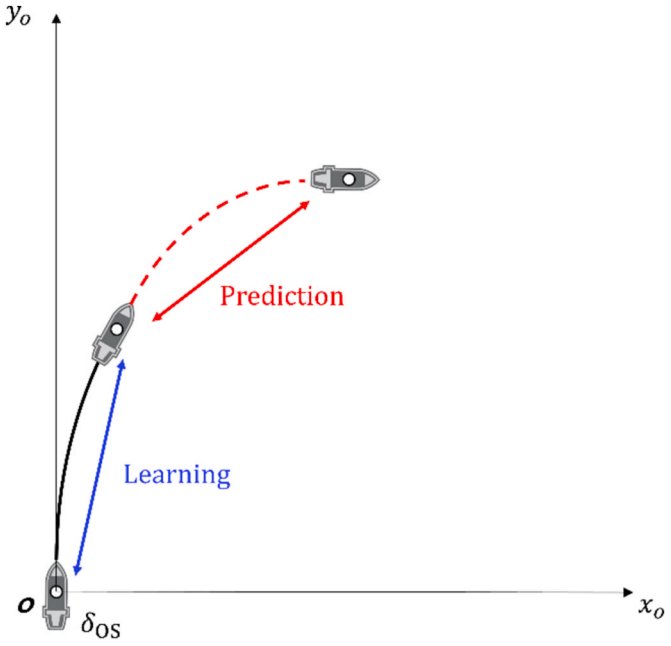


Fig. 8. Ship's trajectory learning and prediction rough illustration.

Table 4
CNN–LSTM architecture and parameter.

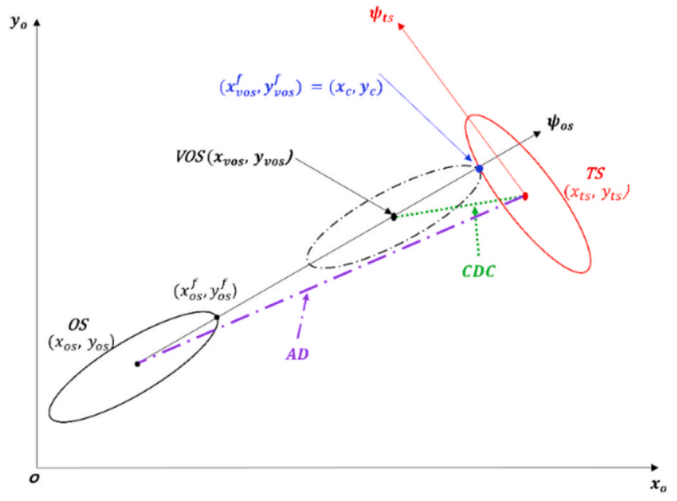
Layer	Output shape	Parameter
Convolutional layer	(None, 56,64)	256
Max pooling layer	(None, 27,64)	0
LSTM	(None, 27,128)	98,816
Flatten	(None, 3456)	0
Dense	(None, 1)	3457

Table 5
CNN–LSTM hyperparameter.

Hyperparameter	Values
LSTM neurons	128
Hidden layer	1
Activation function	tanh
Optimizer	Adam
Batch size	16
Epochs	100
Window size	56
Early stopping (patience)	10–20

2.3.1. *Hydrodynamic forces and moment acting on the hull*

The force acting on the hull is generated by fluid forces in response to the ship's motion. It comprises both linear and nonlinear terms. The



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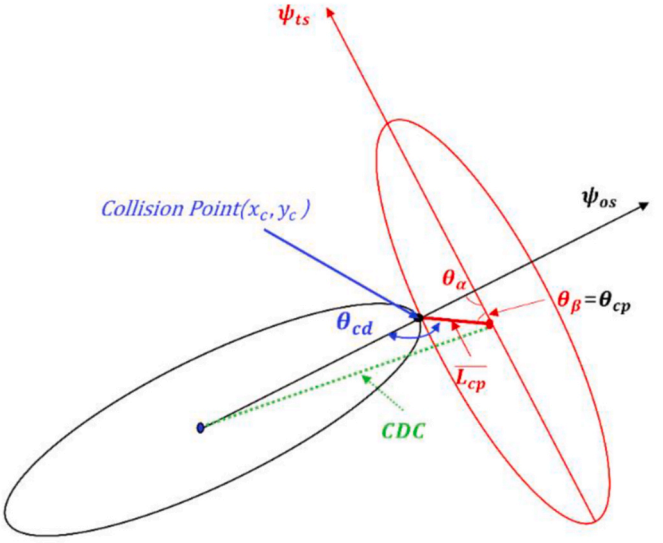


(b)

Fig. 9. Parameters for calculating CDC.

linear terms employ a consistent mathematical expression, whereas the nonlinear terms are characterized by quadratic or cubic expressions.

$$\begin{aligned} X_H &= X_{\dot{u}}\dot{u} + x_{vr}vr + X_{vv}v^2 + X_{rr}r^2 + X(u) \quad Y_H \\ &= Y_{\dot{v}}\dot{v} + Y_{\dot{r}}\dot{r} + Y_{\dot{p}}\dot{p} + Y_r r + Y_v v + Y_{v|v|}v|v| + Y_p p + Y_\varphi \varphi + Y_{v|\varphi|}v|\varphi| + Y_{r|\varphi|}r|\varphi| \\ &\quad -\frac{p}{2}dC_{D0}\left[\int_{-L/2}^{L/2}|v + C_{rY}rx|(v + C_{rY}rx)dx - Lv|v|\right] \quad N_H \\ &= N_{\dot{v}}\dot{v} + N_{\dot{r}}\dot{r} + N_{\dot{p}}\dot{p} + N_r r + N_v v + N_{v|v|}v|v| + N_p p + N_\varphi \varphi + N_{v|\varphi|}v|\varphi| + N_{r|\varphi|}r|\varphi| \\ &\quad -\frac{p}{2}dC_{D0}\left[\int_{-L/2}^{L/2}|v + C_{rN}rx|(v + C_{rN}rx)x - dx\right] \end{aligned} \tag{2}$$

where, In Eq. (2), the various coefficients are tested values from model testing. In Eq. (2) $\frac{p}{2}dC_{D0}[\int_{-L/2}^{L/2}|v + C_{rY}rx|(v + C_{rY}rx)dx - Lv|v|]$ means cross flow drag, $X(u)$ denotes ship's resistance in forward straight motion, φ denotes rolling range, d is ship's draft, x_{vr}, X_{vv} and other similar coefficients denote hydrodynamic derivates.

2.3.2. Longitudinal force produced by a rudder

The rudder is usually located behind the propeller and is therefore affected by the flow velocity generated by the propeller. An appropriate model to represent these perturbations is the Maneuvering Modelling Group (MMG), as shown in Eq. (3).

$$\left.\begin{aligned} X_R &= -(1-t_R)F_N \sin\delta, \\ Y_R &= -(1+a_H)F_N \cos\delta, \\ N_R &= -(x_R + a_H x_N)F_N \cos\delta \end{aligned}\right\} \tag{3}$$

where F_N is the nondimensional normal force on the rudder, t_R and a_H are interaction coefficients, and x_R and x_H denote the positions of the rudder and the acting point of the additional lateral force induced by steering, respectively.

2.3.3. Longitudinal force produced by a propeller

The forces acting on the propeller are also taken from the MMG model. The later components of force and moment are omitted because they are insignificant, and only the longitudinal force is shown as Eq. (4).

$$\left.\begin{aligned} X'_P &= (1-t_P)n^2 D_P^4 K_T(J_P) \Big/ \frac{1}{2}LdU^2, \\ K_T(J_P) &= C_1 + C_2 J_P + C_3 J_P^2, \\ J_P &= U\cos\beta(1-w_P)/nD_P, \end{aligned}\right\} \tag{4}$$

where.

t_P: thrust deduction factor
n: number of a propeller revolution
D_P: diameter of a propeller
$K_T(J_P)$: thrust coefficient
J_P: advance coefficient
C_1, C_2, C_3: constants
w_P: effective wake fraction at propeller position

2.3.4. Wave effect

The motion caused by waves is represented by a sum of regular waves with a very large number of frequencies and directions to represent the actual sea state, but if all of these components are considered, the computation is too large to perform real-time simulation. Therefore, the dynamics model for simulation uses simplified expressions, and in this simulation, it will be considered only three vertical direction motions (heave, roll, and pitch), and simplify them by introducing a force amplification factor as a function of wavelength or wave slope, and then use Eq. (5).

$$z = \zeta_i A_{fac}\cos[\xi + \varepsilon + \omega] \tag{5}$$

where ξ is the phase of the wave at the current location, ε is the phase lag of the response amplitude operator (RAO), and ω is the initial phase according to the force mode. A_{fac} is the amplification factor of KRISO.

2.3.5. Wind effect

For wind forces acting on superstructure of ship on the water surface, only force components acting in the horizontal plane is considered. Because the other direction is relatively small. Coefficient of force and aft component of wind force can be represented as Eq. (6) (Isherwood, 1972).

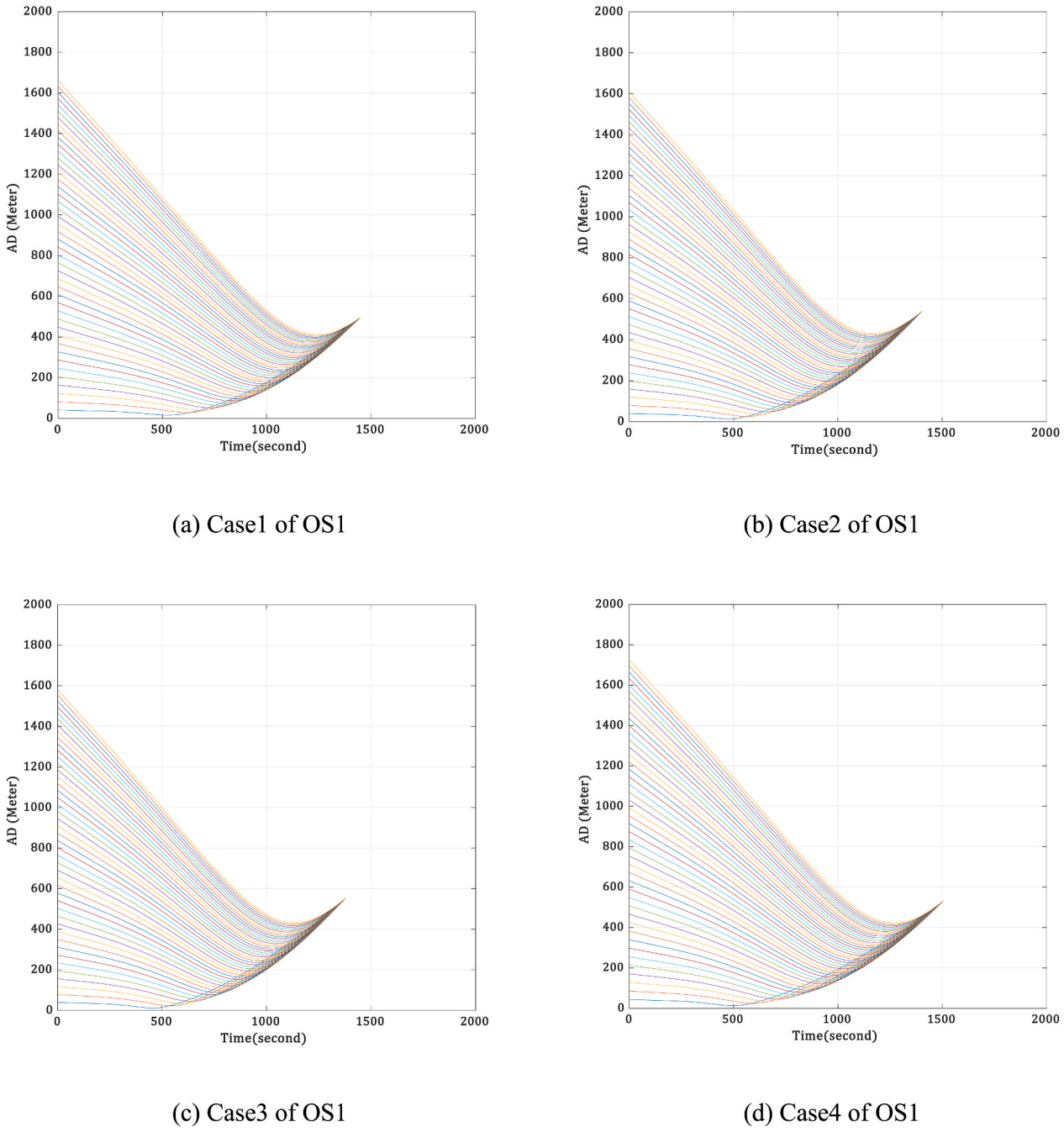


Fig. 10. Actual distance during the simulation.

$$C_X = \frac{F_X}{1/2\,\eta V_R^2 A_T}, \tag{6}$$

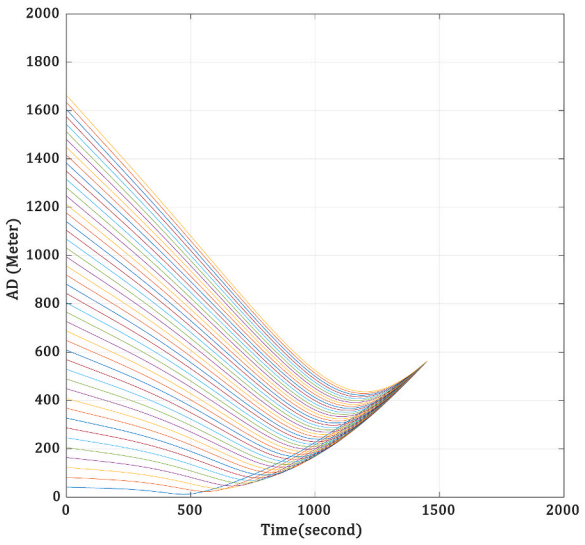
where F_X is the fore and aft components of wind force, η denotes density of air, V_R denotes relative wind speed and A_T denotes transverse projected area of ship.

Eq. (7) shows coefficient of lateral component of wind force.

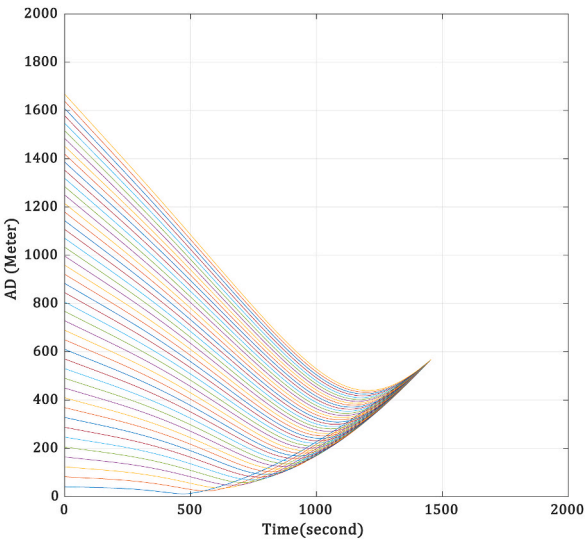
$$C_Y = \frac{F_Y}{1/2\,\eta V_R^2 A_L}, \tag{7}$$

where, F_Y denotes the lateral component of wind force, considered positive when directed away from the wind, and A_L means lateral projected area of ship.

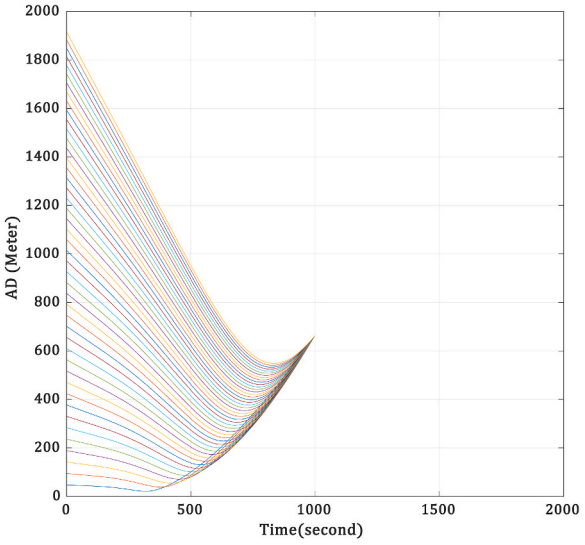
Coefficient of wind-induced yawing moment is shown as Eq. (8)



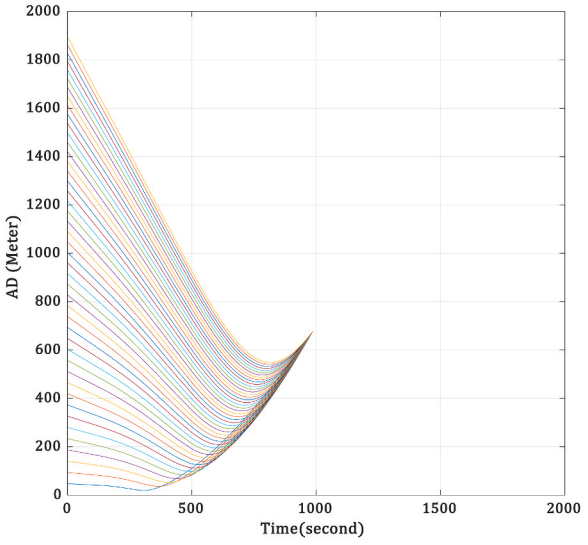
(e) Case5 of OS1



(f) Case6 of OS1



(g) Case1 of OS2



(h) Case2 of OS2

Fig. 10. (*continued*).

$$C_N = \frac{N}{1/2\,\eta V_R^2 A_L L_{OA}}, \tag{8}$$

where N denotes the wind-induced yawing moment about mid-ships overall, considered positive when tending to turn the bow away from the wind, and L_{OA} is the overall length of the ship.

2.3.6. Current effect

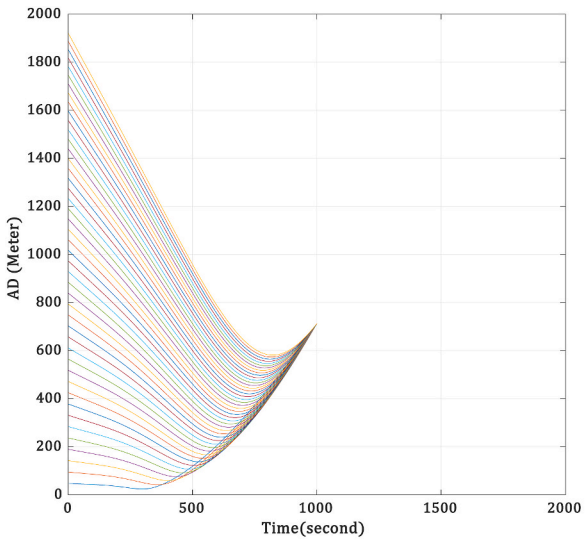
Because inflow velocities on a hull are affected by current, hydrodynamic forces acting on a ship in flow should be included in numerical simulations when a ship is sailing under current.

In order to account for the effect of current, relative speed to the water should be considered. The relative speed components u^* and v^* in x- and y-axes can be represented by including ship speed U, drift angle β and heading angle ψ as shown in Eq. (9).

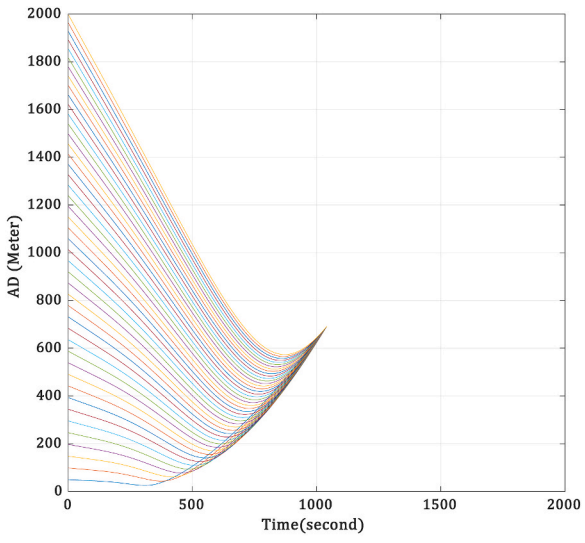
$$\left.\begin{aligned} u^* &= U\cos\beta + V_C\cos(\psi-\alpha), \\ v^* &= -U\sin\beta - V_C\sin(\psi-\alpha). \end{aligned}\right\} \tag{9}$$

2.4. CNN-LSTM

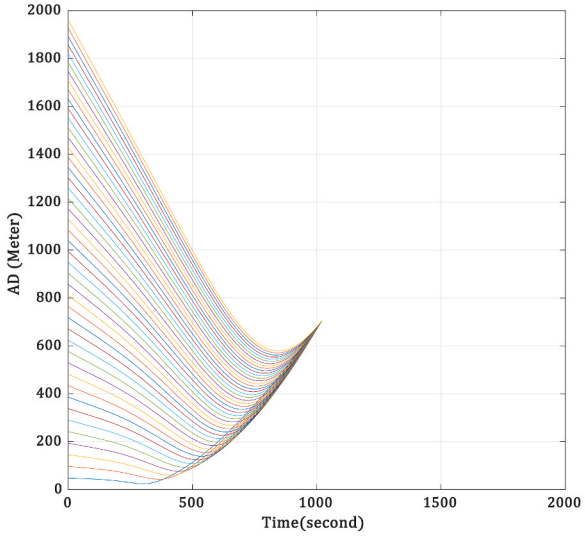
CNN–LSTM is a model that combines the effective data feature



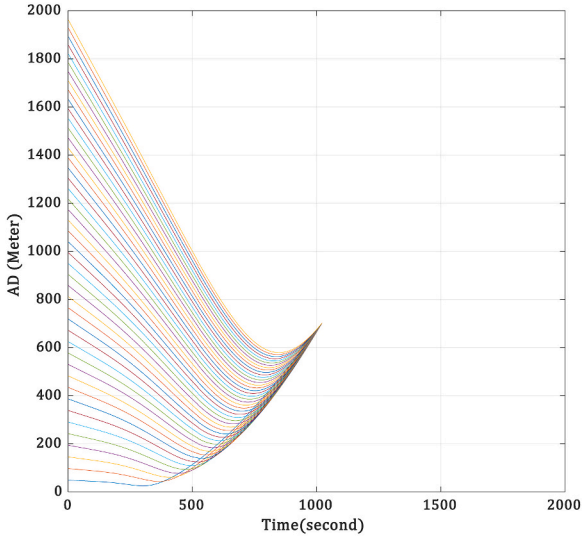
(i) Case3 of OS2



(j) Case4 of OS2



(k) Case5 of OS2



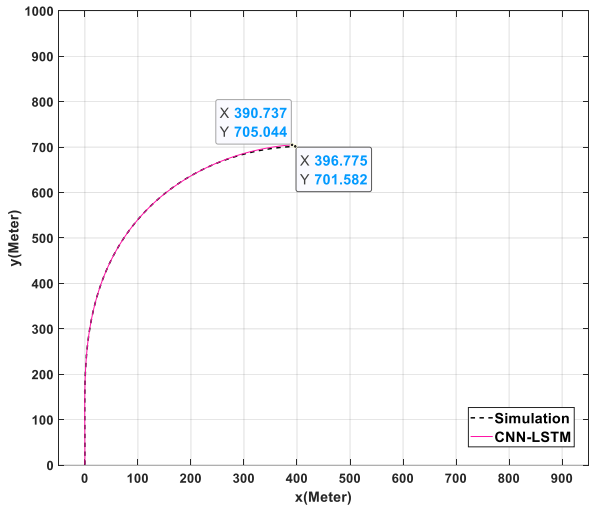
(l) Case6 of OS2

Fig. 10. (*continued*).

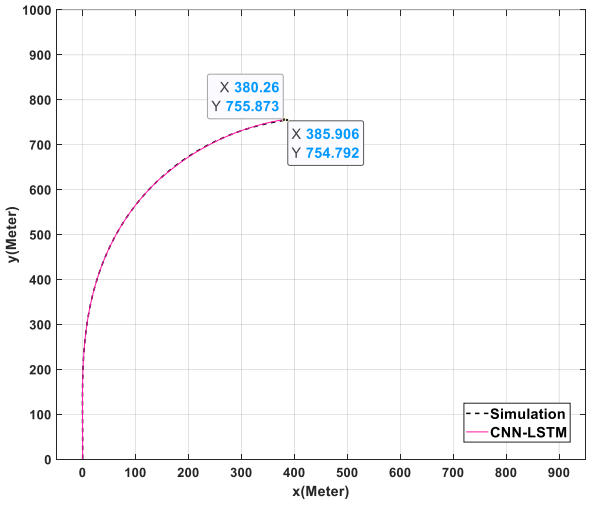
detection ability of CNN with the advantages of LSTM, which excels at processing time series data (Yan et al., 2021). CNN–LSTM has been utilized in various fields such as natural language processing, image recognition and classification (Bhunia et al., 2019; Zhao et al., 2019; Islam et al., 2020), weather prediction (Yan et al., 2021), energy consumption prediction (Kim and Cho, 2019), and gas production prediction (Zha et al., 2022). Therefore, in this study, CNN–LSTM is utilized for ship trajectory prediction.

2.4.1. LSTM

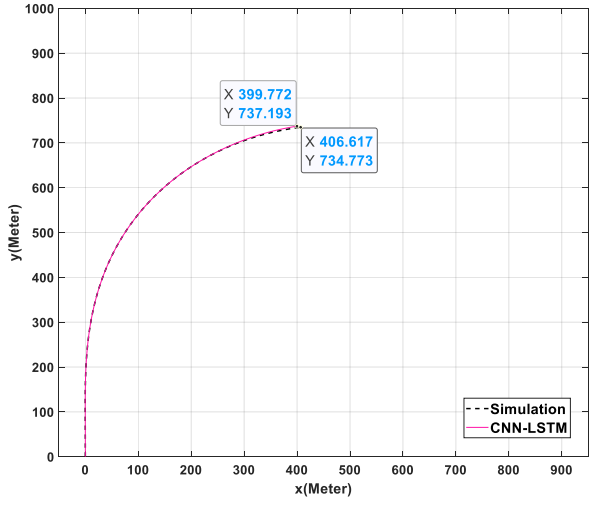
LSTM is a type of recurrent neural network that was developed by Hochreiter and Schmidhuber (1997). LSTM is suitable for predicting



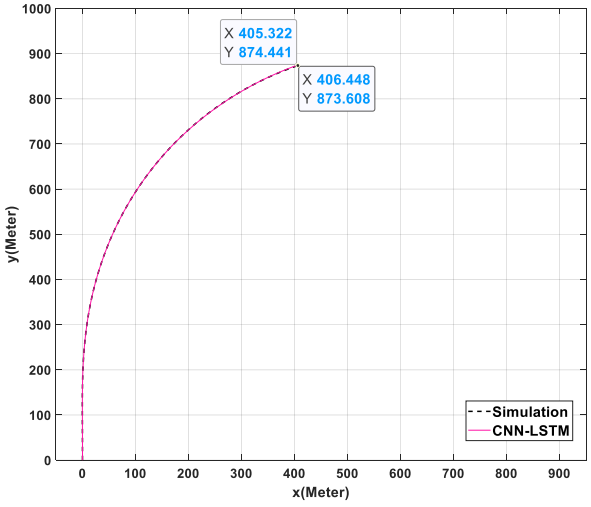
(a) Case1 of OS1



(b) Case2 of OS1



(c) Case3 of OS1



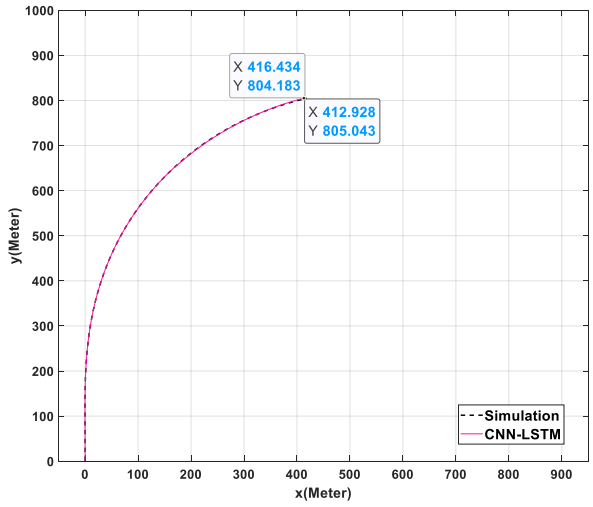
(d) Case4 of OS1

Fig. 11. Compare simulated ship trajectories with ship trajectories from CNN–LSTM.

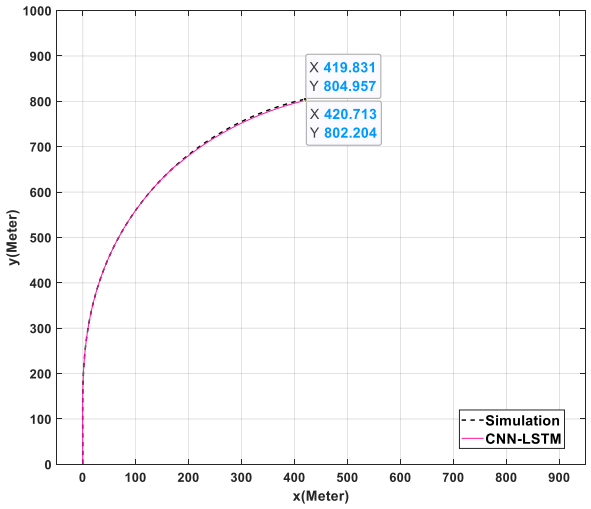
time series data (Abebe et al., 2022), and it has been widely utilized by other researchers to predict ship trajectories (Gao et al., 2021; Zhang et al., 2021; Abebe et al., 2022). Making very precise predictions with LSTM alone is difficult (Gao et al., 2021; Zhang et al., 2021; Abebe et al., 2022). However, in our study, the ship trajectory only needed to be predicted for a short time domain of up to 150 s, so we applied LSTM to predicting ship trajectories because the prediction error would not be significant. Fig. 6 shows a schematic of LSTM, and Eq. (10) describes how it works:

$$\left.\begin{aligned} f_t &= \sigma\left(W_f[x_t, h_{t-1}] + b_f\right) \\ i_t &= \sigma(W_i[x_t, h_{t-1}] + b_i) \\ \breve{C}_t &= \overset{t}{anh}(W_c[x_t, h_{t-1}] + b_c) \\ C_t &= f_t * C_{t-1} + i_t * \breve{C}_t \\ O_t &= \sigma(W_o[x_t, h_{t-1}] + b_o) \\ h_t &= O_t * \tanh(C_t) \end{aligned}\right\}, \tag{10}$$

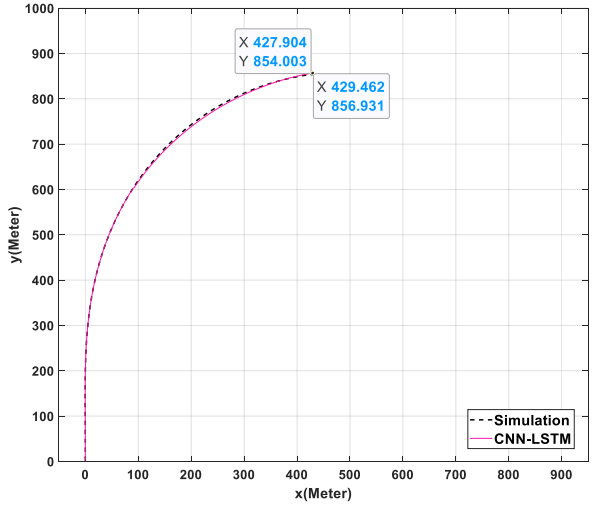
where $W_f, W_i, W_c, and\ W_o$ denote the weights of the forget gate, input gate, output gate, and block input, respectively. $b_f, b_i, b_c, and\ b_o$ denote



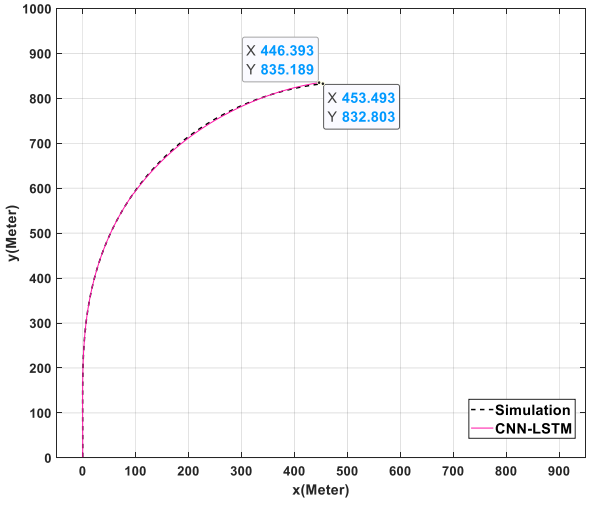
(e) Case5 of OS1



(f) Case6 of OS1



(g) Case1 of OS2



(h) Case2 of OS2

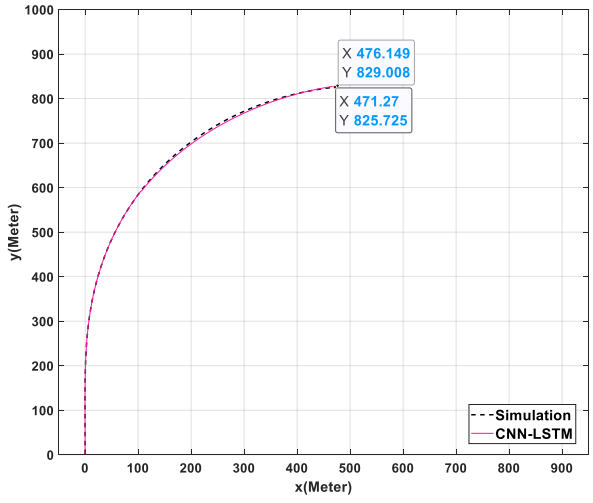
Fig. 11. (*continued*).

the corresponding biases. σ denotes the sigmoid activation function, and tanh is the activation function. C_t denotes the current memory state, C_{t-1} is the previous memory state, O_t is the current output, x_t is the current input, h_t is the current hidden state, and h_{t-1} is the previous hidden state. The forget gate determines how much past data to forget, and the input gate determines how much to utilize x_t through h_{t-1}. The cell state determines how much information from the input gate and forget gate to combine. The output gate utilizes data from the hidden and input gates and the information from the cell state to produce the output.

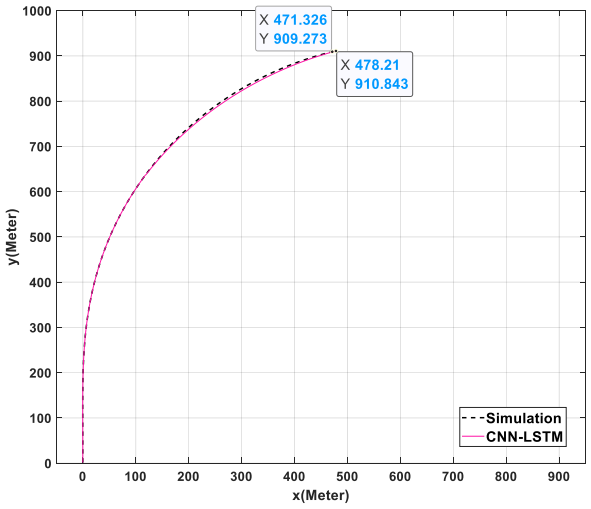
2.4.2. CNN

CNN has shown outstanding results in image recognition, natural language processing, speech recognition, and time series data processing (Hamad et al., 2020). The structure of CNN consists of input, convolutional layer, pooling layer, and fully connected layer.

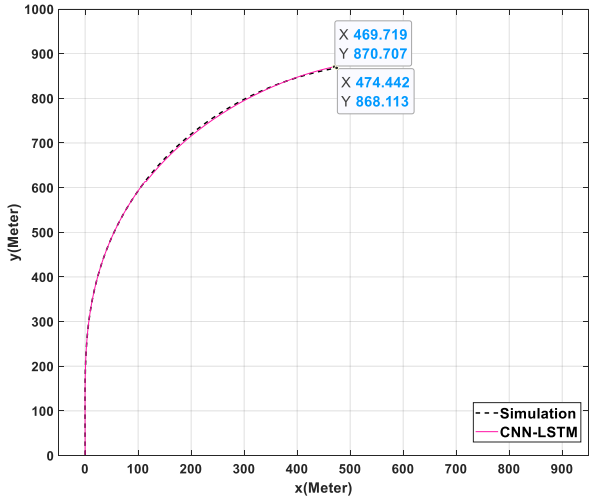
The convolutional layer is the most important layer in a CNN and detects specific features or patterns in the data (Yan et al., 2021). In this layer, after the convolutional computation, an activation function is used to introduce the nonlinearity needed to learn complex patterns, which is called the relu function in this study (Zha et al., 2022).



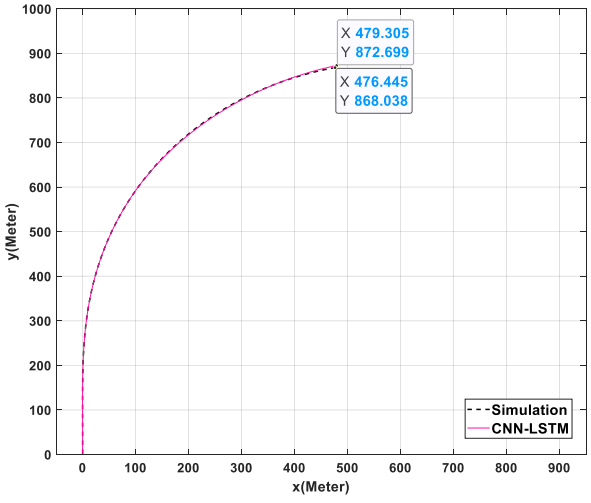
(i) Case3 of OS2



(j) Case4 of OS2



(k) Case5 of OS2



(l) Case6 of OS2

Fig. 11. (*continued*).

The pooling layer performs down-sampling to reduce the amount of computation and increase efficiency. There are two pooling methods, average pooling and maximum pooling, and this study uses maximum pooling. Both methods extract the maximum or average value from the pooling region, but max pooling is better for finding the strongest features or patterns within the given data (Wang et al., 2016; Zha et al., 2022). The features passed through the pooling layer are passed to the fully connected layer, and the output value of the CNN is calculated in the fully connected layer. (Swapna et al., 2018). However, in CNN–LSTM, the fully connected layer is computed after the LSTM. This calculation process is represented by Eq. (11).

$$Y = Relu\left(Relu\left(\bullet\bullet\bullet Relu\left(b_j^k + \sum_{i=1}^{N_{k-1}} Conv\left(x_i^{k-1}, \omega_{ij}^{k-1}\right)\right)\right)\right) \tag{11}$$

where, b_j^k is the threshold value of layer k, N_k is set of input features, x_i^{k-1} is input value of layer k-1, ω_{ij}^{k-1} is convolution kernel. The Y features are

Table 6
CNN–LSTM maximum values of training result.

	Epoch	Loss	MAE	Value loss	Value MAE
Case1 of OS1	22	3.7629e−07	4.8762e−04	5.1798e−05	0.0054
Case1 of OS2	45	2.2035e−07	3.7953e−04	4.8331e−05	0.0046
Case2 of OS1	30	2.9090e−07	4.3198e−04	6.0504e−05	0.0054
Case2 of OS2	40	5.1583e−07	5.7694e−04	4.2681e−05	0.0060
Case3 of OS1	25	2.8578e−07	4.2761e−04	3.7167e−05	0.0042
Case3 of OS2	38	5.6565e−07	6.1606e−04	3.3030e−05	0.0040
Case4 of OS1	38	6.1809e−07	6.3048e−04	5.2020e−05	0.0053
Case4 of OS2	52	2.9676e−07	4.2405e−04	1.2571e−05	0.0026
Case5 of OS1	35	3.2341e−07	4.3042e−04	4.3945e−05	0.0046
Case5 of OS2	60	2.1956e−06	0.0011	1.7969e−05	0.0037
Case6 of OS1	32	1.5960e−07	3.1058e−04	2.4177e−05	0.0033
Case6 of OS2	48	8.7548e−08	2.3327e−04	2.7508e−05	0.0032

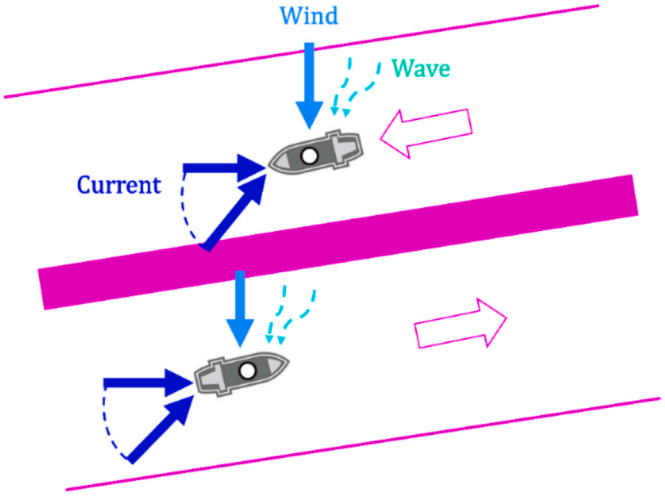


Fig. 12. The direction of the sea condition acting on the ship in this study.

Table 7
Comparison between λ and λ_o.

	Drift distance (d_{df})	
	OS1	OS2
Case 1	0.32L	0.15L
Case 2	0.16L	0.07L
Case 3	0.21L	0.11L
Case 4	0.38L	0.18L
Case 5	0.12L	0.05L
Case 6	0.12L	0.04L

derived from the input data, and these extracted features are then fed into the LSTM for prediction.

The architecture of the combined LSTM and CNN described earlier and the architecture designed in this study is shown in Fig. 7 and Table 4.

As shown in Table 4, the model starts with a Convolutional layer using 64 filters and a kernel size of 3, activated by the Relu function. Subsequently, a MaxPooling layer is applied with a window size of 2, reducing dimensions. An LSTM Layer follows with 128 units and a tanh activation function, capturing temporal information while retaining sequences. The Flattening layer transforms LSTM outputs into a 2D format. Finally, a Dense Layer generates regression values with a single output using linear activation.

As shown in Fig. 8, we used CNN–LSTM to learn the simulated ship trajectories and predict the last 600 trajectory data.

Table 5 presents the LSTM architecture. The hyperparameters were set to 128 neurons, Adam as the optimizer, and the mean absolute error (MAE) as the loss function. We also terminated training early when the validation loss value did not improve more than 10–20 times. The final prediction was displayed as the original value by inverse transformation of the scaled data.

The training data set (Window size) learns the x- and y-coordinates of the OSs from 0 to 55 s (56 set) and then predicts the x- and y-coordinates at 56 s. It then learns the data of the next coordinates (1–56 s) to predict the x- and y-coordinates at 57 s. A window size of 56 was found to maximize the prediction accuracy. So, in this way, 600 sets of final OS coordinates were predicted (corresponding to the last 60.0 s of trajectory).

2.5. Computed distance at collision

A collision assessment (CA) model was developed to calculate the timing for ship action. The ship domain can be defined as the safety margin between ships for collision avoidance. Therefore, it is difficult to identify for specific collision situations. The distance to closest point of approach (DCPA) is a representative CA technique, but it has difficulty with identifying the collision distance between ships and specific time of collision because it is calculated based on the position of the Global Positioning System antenna on polyhedral ships (Lee and Furukawa, 2022). A member of our group previously proposed the CDC (Lee and Furukawa, 2022), which considers the length and width of two ships, the heading angle between them, and the collision angle to calculate all possible collision distances. In this study, we applied the CDC in the CA model, although we shortened the time interval from the original 0.5 s (Lee and Furukawa, 2022) to 0.1 s for a more detailed calculation.

The CDC can be used to evaluate ship-to-ship collisions based on the extended centerline of the OS and the ellipse equation of the TS and the length and width of ship. Fig. 9 show how the CDC is calculated.

The black ellipse is the OS, the black dotted ellipse is the virtual OS (VOS), and the red ellipse is the RS. In an Earth-fixed coordinate system, the reference point of the OS is denoted by x_{os}, y_{os}, the reference point of the TS is denoted by x_{ts}, y_{ts}, and their headings are denoted by ψ_{os} and ψ_{ts}, respectively. The reference point of the VOS is denoted by x_{vos}, y_{vos}, and the coordinates of the OS and VOS headings are denoted by x^f_{os}, y^f_{os}, x^f_{vos}, and y^f_{vos}.

The coordinates of the point where the two ships collide are x_c, y_c.

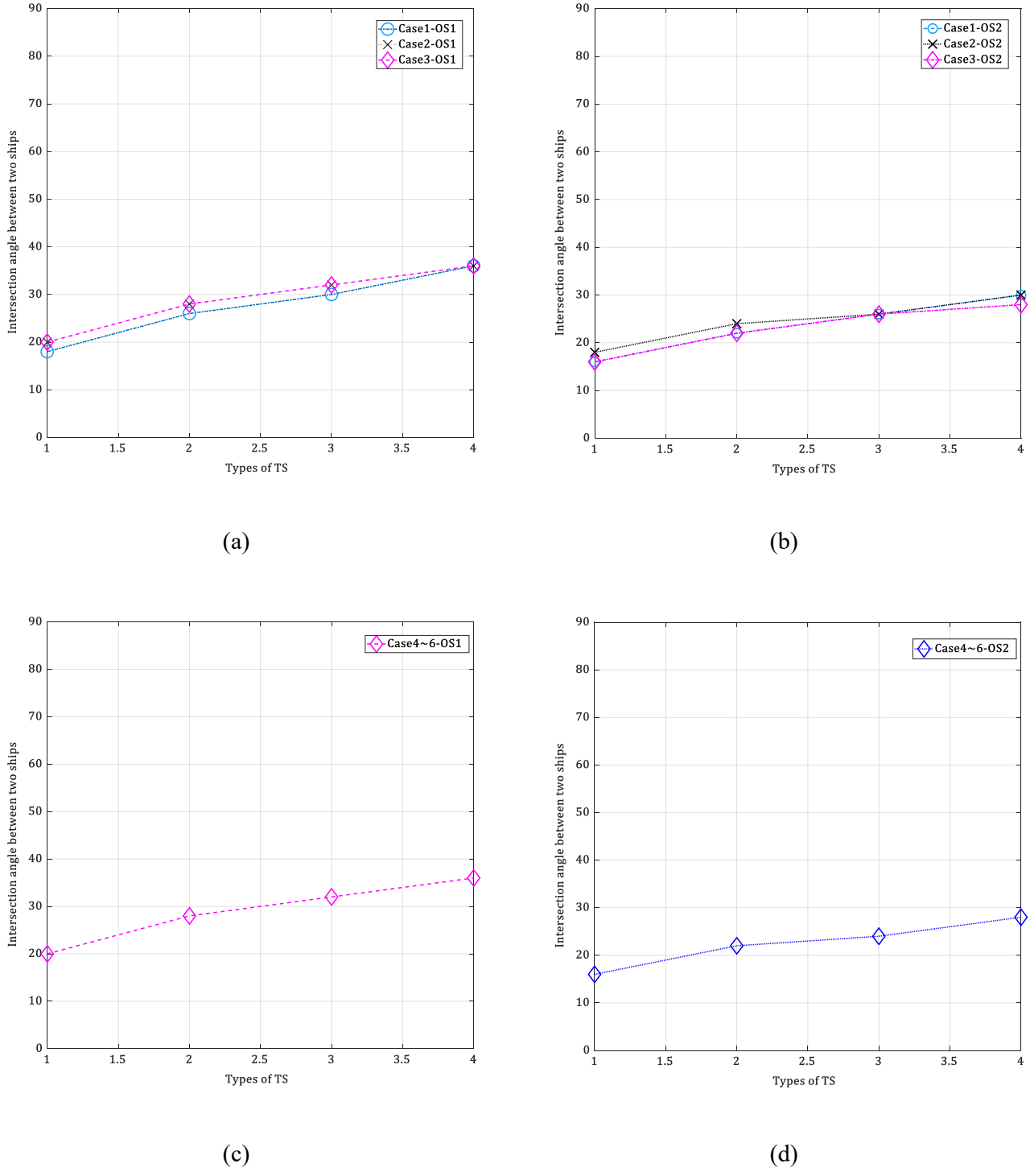


Fig. 13. Result of CA

Then, the CDC is defined as the distance between the two ships at the time of collision, and AD is the actual distance between the two ships. To calculate the CDC, the OS is moved to a virtual position without changing its heading so that it is in contact with the RS. The heading of the OS and the ellipse of the TS will then be tangential (i.e., the extended centerline of the OS will be tangential to the ellipse of the TS). The ellipse equation for the TS is given by

$$\frac{\{(y_o-y_{ts})\cos\psi_{ts}+(x_o-x_{ts})\sin\psi_{ts}\}^2}{(L/2)^2}+\frac{\{-(y_o-y_{ts})\sin\psi_{ts}+(x_o-x_{ts})\cos\psi_{ts}\}^2}{(B/2)^2}=1 \tag{12}$$

The extended centerline of the OS is given by

$$y_o=(x_o-x_{os})\tan\psi_{os}+y_{os} \tag{13}$$

Table 8
Collision assessment results (θ_{ic}, t_c, and d_c).

		Case 1		Case 2		Case 3		Case 4		Case 5		Case 6	
		OS1	OS2	OS1	OS2	OS1	OS2	OS1	OS2	OS1	OS2	OS1	OS2
TS1	θ_{ic}	20 °	16 °	20 °	18 °	20 °	16 °	20 °	16 °	20 °	16 °	20 °	16 °
	t_c	73.7 s	49.7 s	76.8 s	51.2 s	73.4 s	49.3 s	82.3 s	51.9 s	77.9 s	50.5 s	78.2 s	50.6 s
	d_c	122 m	144 m	122 m	145 m	120 m	142 m	122 m	145 m	122 m	144 m	123 m	144 m
TS2	θ_{ic}	28 °	22 °	28 °	24 °	28 °	22 °	28 °	22 °	28 °	22 °	28 °	22 °
	t_c	84.8 s	56.7 s	88.0 s	57.6 s	83.9 s	56.0 s	94.2 s	59.3 s	89.5 s	57.5 s	90.0 s	57.7 s
	d_c	159 m	187 m	158 m	182 m	154 m	182 m	156 m	185 m	158 m	184 m	159 m	185 m
TS3	θ_{ic}	32 °	26 °	32 °	26 °	32 °	26 °	32 °	24 °	32 °	24 °	32 °	24 °
	t_c	90.2 s	58.4 s	93.4 s	59.1 s	89.2 s	60.9 s	99.7 s	61.0 s	95.0 s	59.3 s	95.4 s	59.5 s
	d_c	182 m	189 m	179 m	196 m	174 m	212 m	175 m	200 m	178 m	198 m	180 m	199 m
TS4	θ_{ic}	36 °	28 °	36 °	30 °	36 °	28 °	36 °	28 °	36 °	28°	36 °	28 °
	t_c	95.1 s	62.6 s	98.2 s	63.0 s	93.7 s	61.8 s	104.4 s	65.3 s	99.7 s	63.5 s	100.4 s	63.7 s
	d_c	208 m	234 m	203 m	223 m	196 m	226 m	195 m	229 m	201 m	228 m	204 m	229 m

Table 9
Collision assessment results under calm conditions.

	$\theta_{ic}(t_c)$			
	TS1	TS2	TS3	TS4
OS1	20 ° (77.6 s)	28 ° (89.0 s)	32 ° (94.5 s)	36 ° (99.2 s)
OS2	16 ° (50.7 s)	22 ° (58.0 s)	24 ° (59.7 s)	28° (64.0 s)

Substituting Eq. (13) into Eq. (12) obtains

$$Ax_o^2 + 2Bx_o + C = 0 \tag{14}$$

Then, x_o can be obtained

$$x_o = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \quad (x_o \in x_c) \tag{15}$$

We then define the following terms:

$$\left.\begin{aligned} L_{cp} &= \sqrt{(x_c - x_{ts})^2 + (y_c - y_{ts})^2}, \\ \theta_{cd} &= \psi_{os} + (2\pi - \psi_{ts}) + \tan^{-1}\left(\frac{x_c - x_{ts}}{y_c - y_{ts}}\right). \end{aligned}\right\} \tag{16}$$

These terms can then be used to calculate the CDC:

$$CDC = \sqrt{\left(\frac{L_{OS}}{2}\right)^2 + (L_{cp})^2 - 2\left(\frac{L_{OS}}{2}\right)L_{cp}\cos\theta_{cd}} \tag{17}$$

The AD can be obtained by using the reference points of the OS and RS:

$$AD = \sqrt{(x_{os} - x_{ts})^2 + (y_{os} - y_{ts})^2} \tag{18}$$

3. Results

We applied the CDC to the CA of ship trajectories predicted by CNN-LSTM (λ_{cl}). First, we determined the AD between the two ships to evaluate the accuracy of λ_{cl}. Then, the CA results were used to determine the timing of ship actions in different cases. Finally, we verified the reliability of the CA results.

3.1. Actual distance between two ships

The initial distance between two ships (d_i) depends on the simulation time (t_s) and the intersection heading angle (θ_{ot}). t_s is determined by $t_{\psi_{os}=90°}$, which depends on sea conditions such as the wind, wave, and current. Meanwhile, a larger θ_{ot} increases d_i at $t = 0$. According to Table 1, OS2 was faster than OS1, so d_i should be longer for OS2. However, it is hard to confirm if the AD is calculated correctly because there are many data to confirm. To confirm the correctness of the calculation, we instead visualized the AD in the OSs and TSs trajectory scenarios. Fig. 10 show the AD during the simulation of each case.

The horizontal axis represents the time, and the vertical axis represents the AD in meters. AD is the distance between the predicted trajectory (λ_{cl}) data by CNN–LSTM and the TS trajectory. The curved lines represent the AD over t_s, where the bottom curved line is the AD at $\theta_{ot} = 2°$ and the top curved line is the AD at $\theta_{ot} = 90°$. Because θ_{ot} was adjusted in intervals of 2°, 45 ADs are shown. In each case, the AD decreased with time and then increased. OS1 had a longer t_s because $t_{\psi_{os}=90°}$ was longer.

In this scenario, the projected AD between the two ships is expected to decrease and then increase during the simulation. If we know the closest AD, we can predict the time of a collision. Fig. 10 shows the expected linearity of the ADs in this scenario, indicating that the AD calculation is appropriate. However, the accuracy of λ_{cl} must also be confirmed. This confirmation will be described in the next section.

3.2. Comparison between the predicted and simulated ship trajectories

Fig. 11 shows the comparison of λ_{cl} with λ_s.

The horizontal axis is the x-coordinate, and the vertical axis is the y-coordinate. The prediction error was from 0.006L to 0.025L, where L is the ship length. The small error is because t_s was ~2 min at most, and the time interval was 0.1s. Table 6 presents the CNN–LSTM training results.

The loss and MAE were used as evaluation metrics of how well CNN−LSTM predicted the training dataset, while the value loss and value MAE were used as evaluation metrics of how well CNN−LSTM predicted the validation dataset. After training on multiple datasets, CNN−LSTM demonstrated a relatively high level of predictive accuracy with a maximum MAE of 0.0060. Here, we aimed to analyze the effect of the sea conditions on ship's trajectory. Fig. 12 shows the sea conditions applied in this study.

Table 7 presents the drift distance (d_{df}), which is defined as the difference in the final coordinates at the end of t_s between λ and λ_o.

The largest d_{df} was observed for Case 4, especially for OS1. This is probably because OS2 had a greater L and ship speed (v), so it was less affected by the sea conditions. In addition, t_s was short, which helps explain why d_{df} was generally not large.

3.3. Collision assessment results

The CA between the OSs and TSs was performed in six cases using the

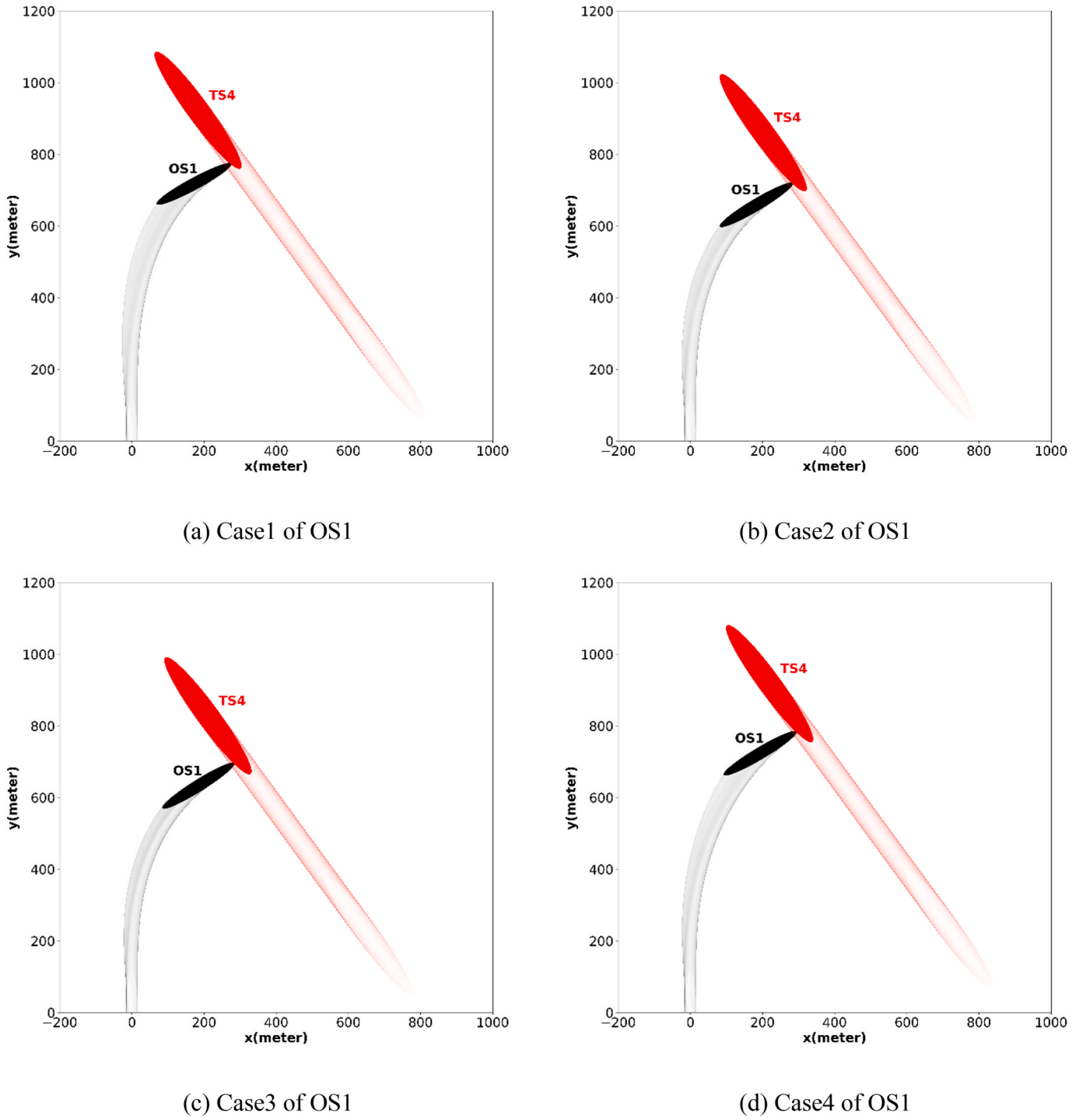
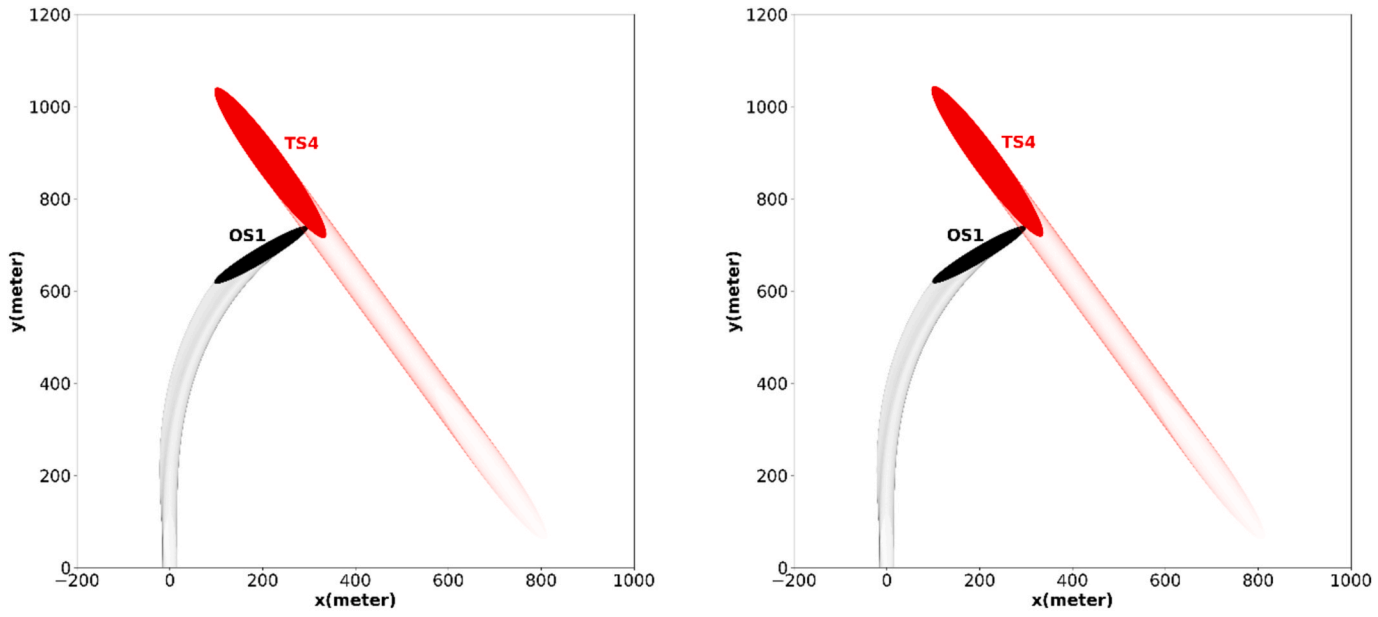


Fig. 14. Compare simulated ship trajectories with ship trajectories from CNN–LSTM.

above-presented CDC. Fig. 13 shows the CA results for OS1 and OS2 in the six cases.

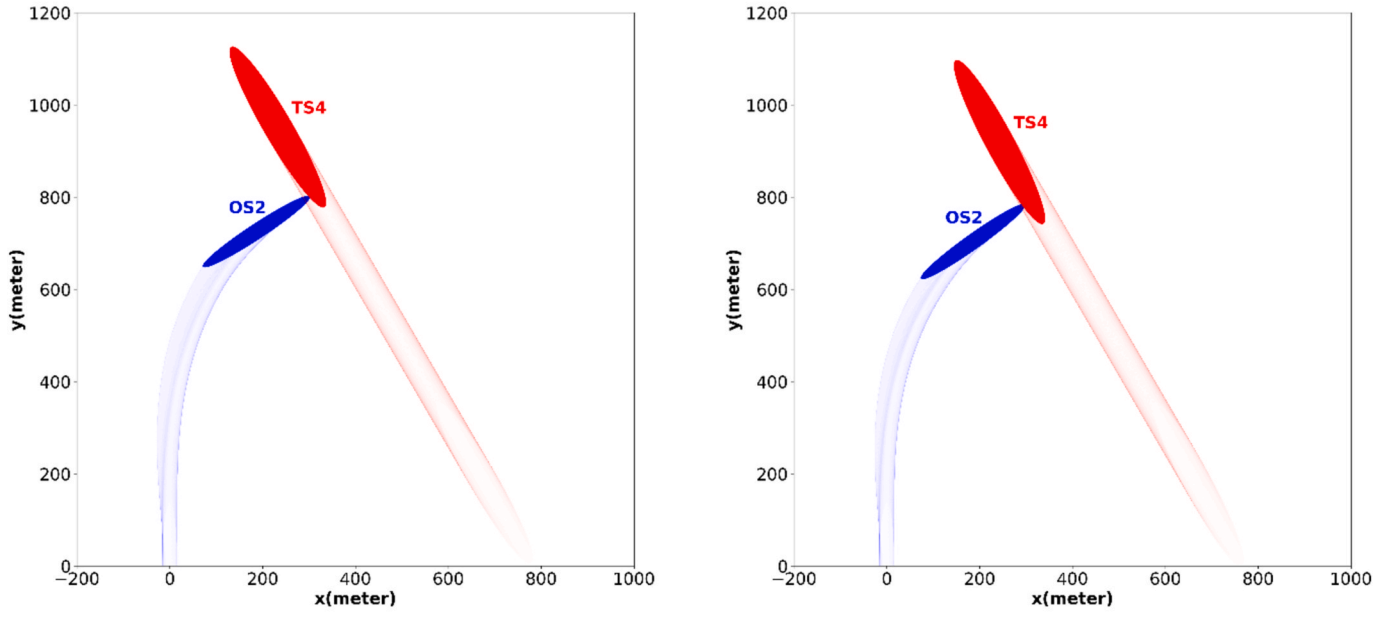
The horizontal axis represents the type of TS, the vertical axis is θ_{ot}, and the marks indicate the intersection heading angle at which collision cannot be avoided between the two ships (θ_{ic}). θ_{ic} increased linearly with L of TS, and OS2 had a smaller max θ_{ic} than OS1, which we attributed it to having a shorter $t_{\psi_{os}=90^\circ}$. For OS1, max θ_{ic} was 36° at TS4 = 400 m, and the different cases did not affect θ_{ic} although they did affect the time of collision. For OS2, max θ_{ic} was 30° when TS = 400 m in Cases 1 and 2. Depending on the size of the TS, $\Delta\theta_{ic}$ varied by 16° for OS1 and by 12° for OS2. Because the CA was performed for four types of TSs, θ_{ic} for TSs of other sizes cannot be known exactly, but it can be approximated.

As shown in Fig. 13, the CA results were affected by the maneuverability of the ship. OS2 is only approximately 30 m longer than OS1 but



(e) Case5 of OS1

(f) Case6 of OS1



(g) Case1 of OS2

(h) Case2 of OS2

Fig. 14. (*continued*).

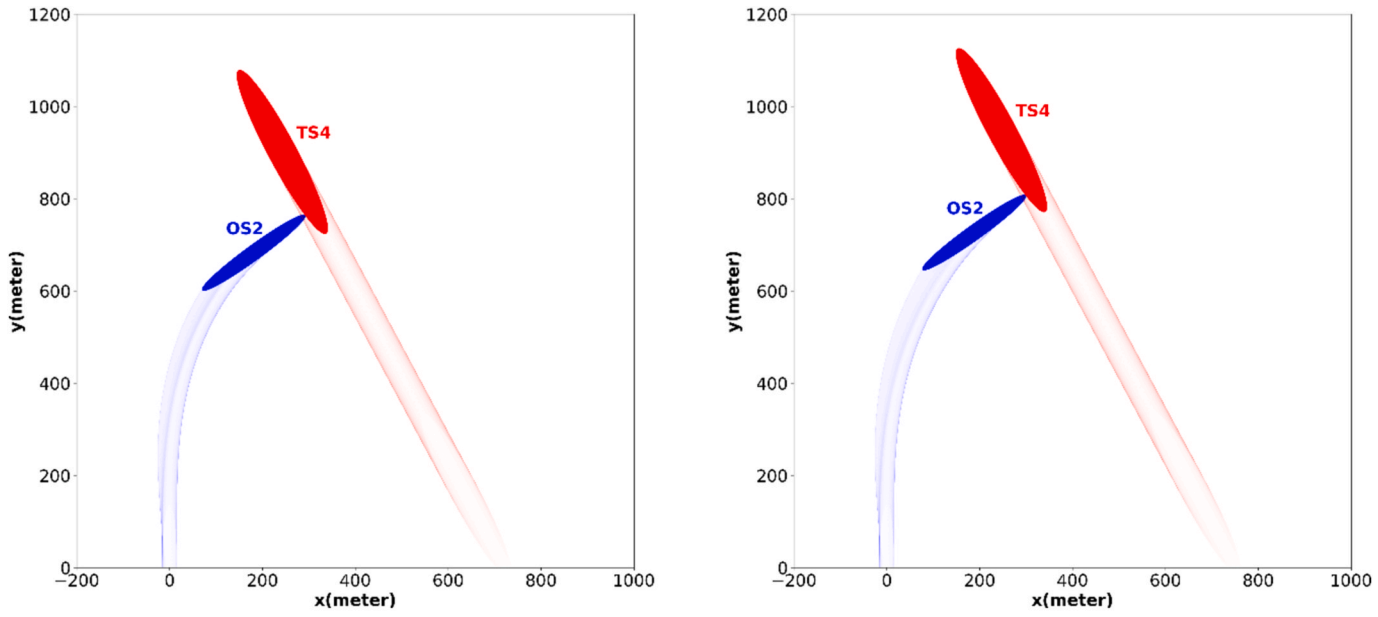
the maximum θ_{ot} between OS1 and OS2 differs by approximately 6°. It was presumed that maneuverability more strongly influences the maximum θ_{ot} than the length and width of the ship.

Table 8 presents θ_{ic}, the collision time (t_c), and distance at collision (d_c) by case and TS type.

OS1 had a larger θ_{ic} than OS2, which indicates that θ_{ic} is strongly affected by $t_{\psi_{os}=90^\circ}$. It was somewhat difficult to analyze which sea conditions (i.e., wind, waves, and current) played the biggest role because the sea conditions had different units of velocity and directions.

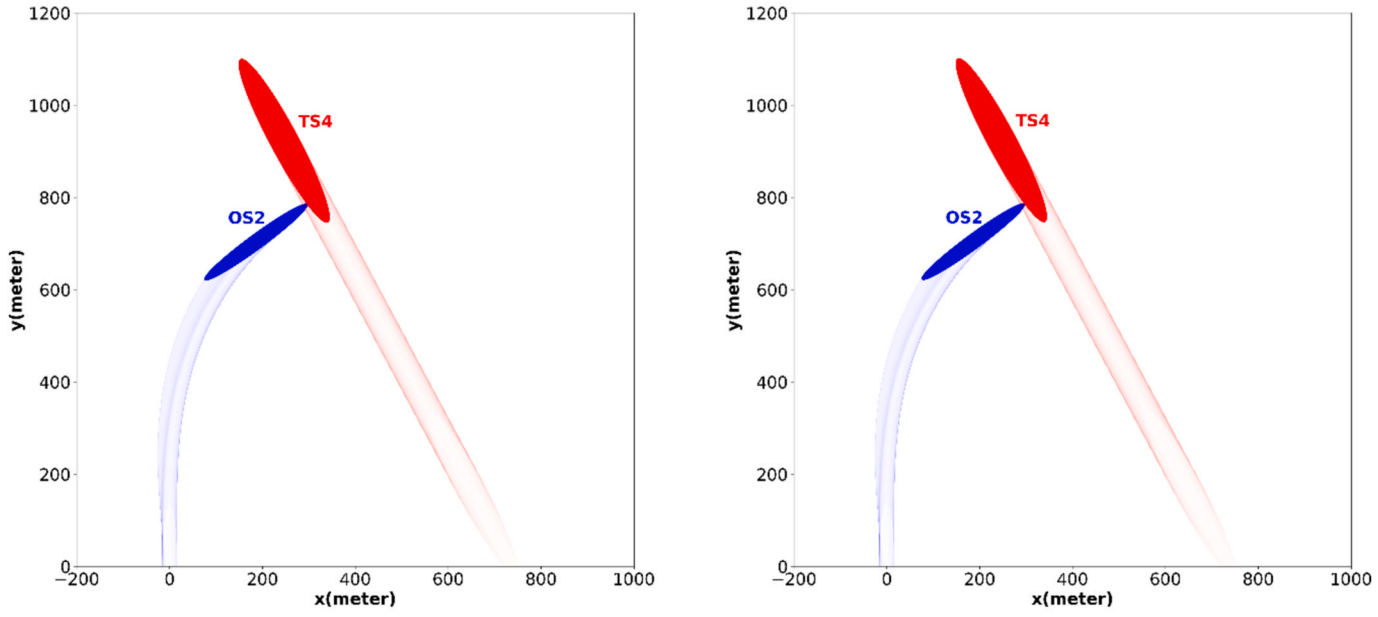
However, we could observe the general trend. In all six OS1 cases, the θ_{ic} value was identical and the t_c and d_c values (which are more detailed and various than θ_{ic}) little depended on the sea condition. The maximum Δd_c was 7 m and Δt_c never exceeded 10 s.

In the OS2 cases, the θ_{ic} sometimes changed by approximately 2° depending on the sea condition. Despite this apparently large difference, the maximum Δd_c and Δt_c never exceeded 13 m and 5 s, respectively.



(i) Case3 of OS2

(j) Case4 of OS2



(k) Case5 of OS2

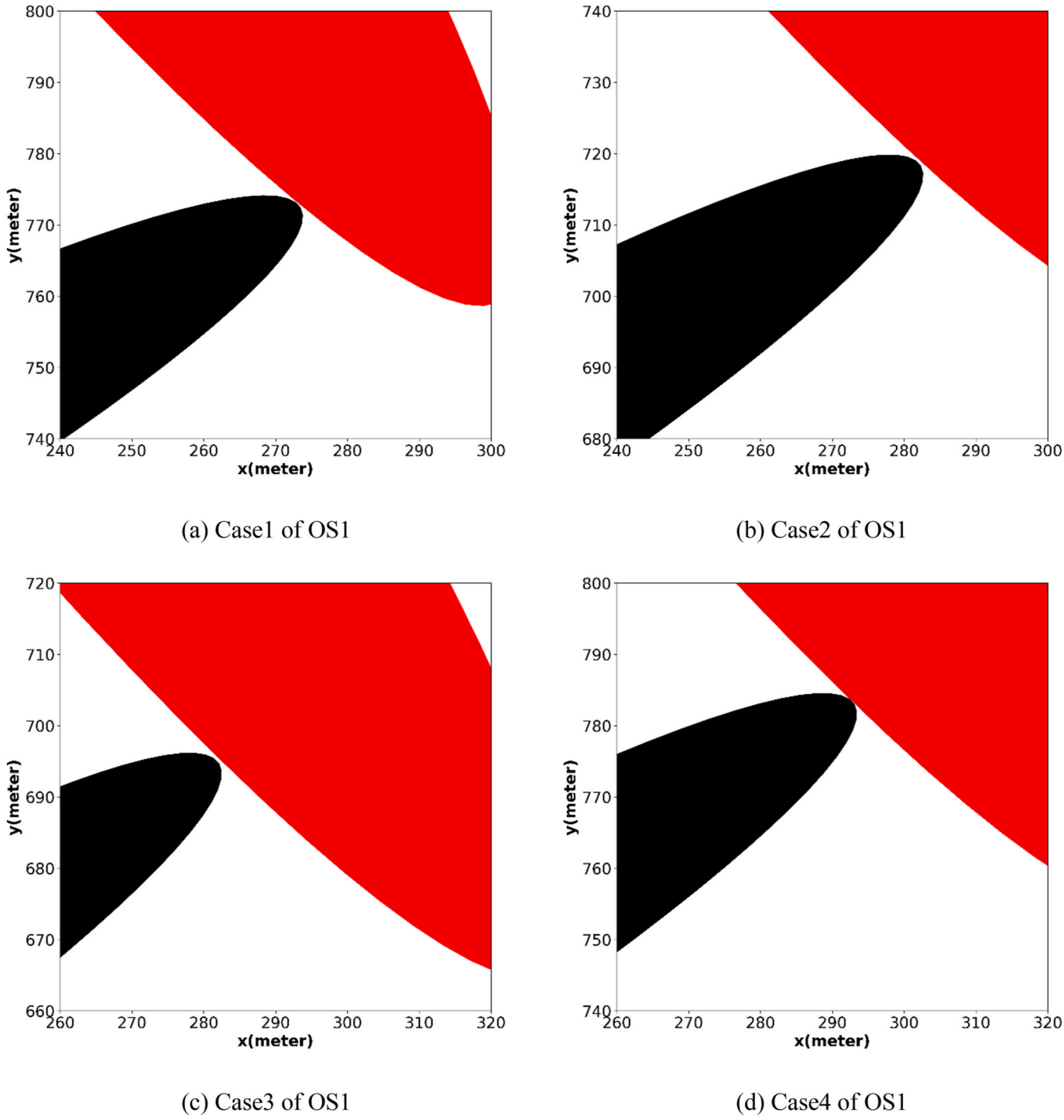
(l) Case6 of OS2

Fig. 14. (*continued*).

The larger d_c of OS2 than of OS1 can be explained by the larger length and width of OS2. Meanwhile, the smaller t_c of OS2 than of OS1 is attributed to the faster speed of OS2 than of OS1.

In a similar study, Lee and Furukawa (2022) used the KVLCC2 model for the OS and TS. In contrast, we used a bulk carrier and container carrier. They obtained $\max \theta_{ic}$ of 34° for KVLCC2, but they did not consider sea conditions. Therefore, to understand the effect of the sea conditions, we conducted CA using λ_o, and the results are presented in Table 9.

The sea conditions did not appear to significantly affect θ_{ic}, which indicates that they have a smaller effect on the CA results than we expected. This may be because the ships are sailing for a very short time in the different cases, and the waves, current, and wind do not act in the same direction, so the forces acting on the ship are distributed as shown in Fig. 12. The results may differ in extremely bad weather, but this was beyond the scope of this study. Considering the results of Lee and



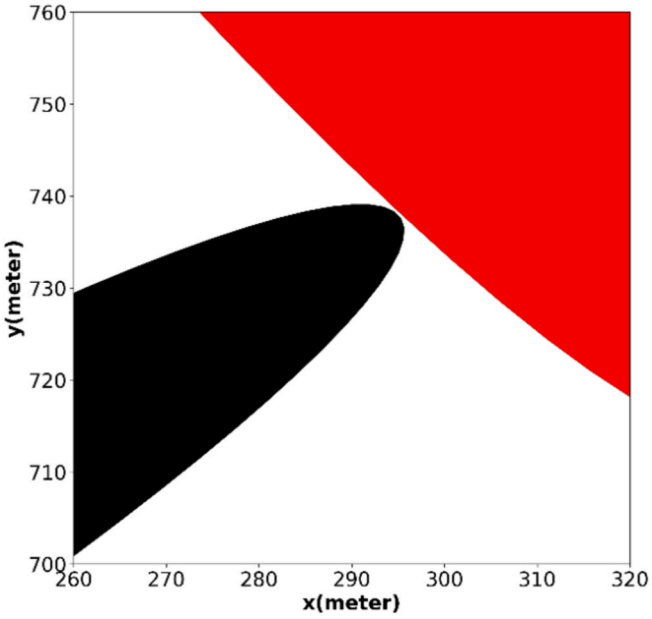
(a) Case1 of OS1

(b) Case2 of OS1

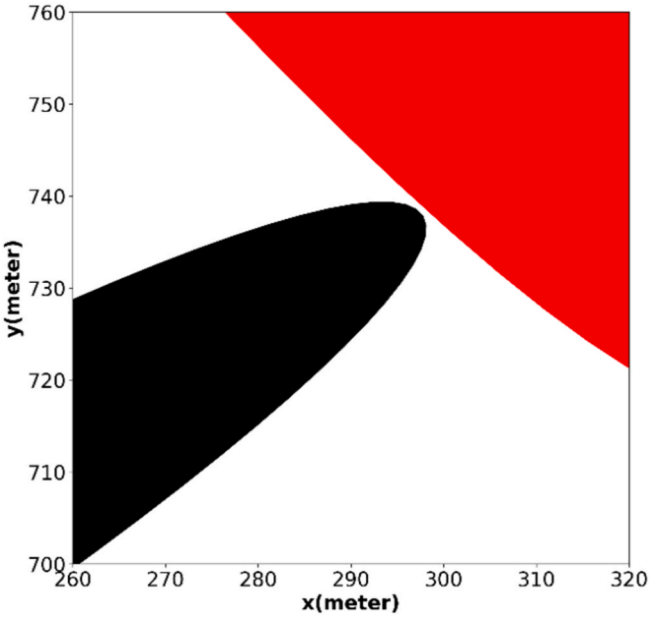
(c) Case3 of OS1

(d) Case4 of OS1

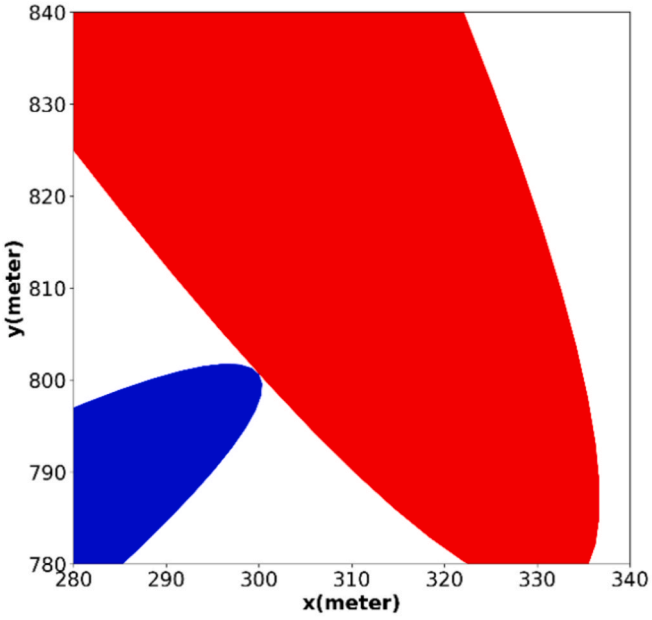
Fig. 15. Two ships 0.1 s before collision.



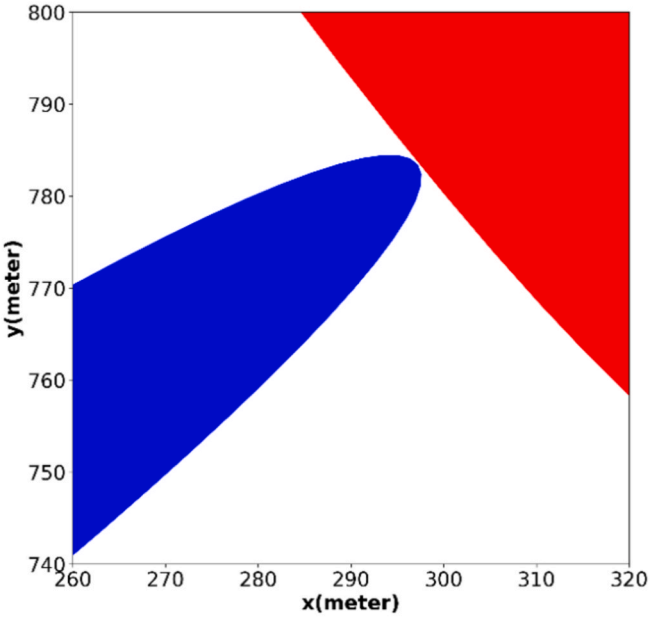
(e) Case5 of OS1



(f) Case6 of OS1

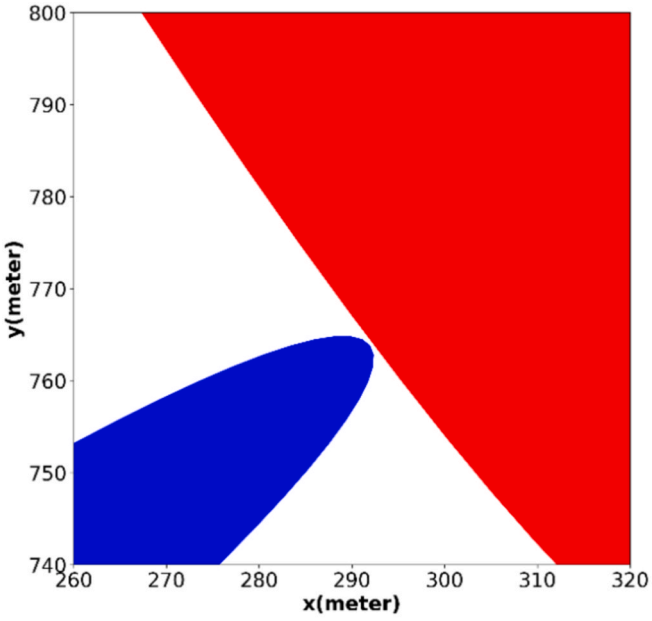


(g) Case1 of OS2

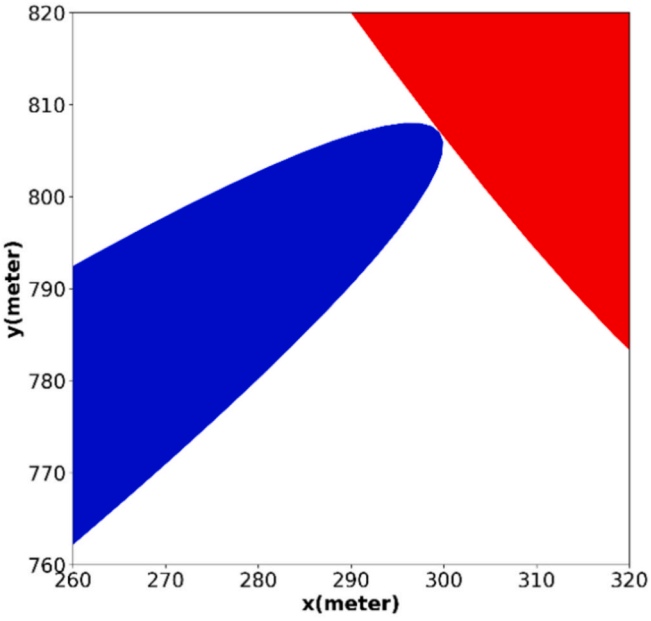


(h) Case2 of OS2

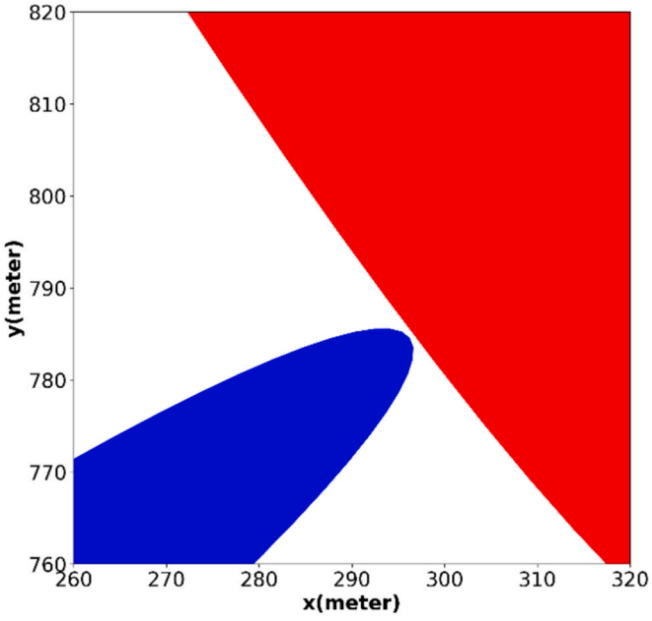
Fig. 15. (*continued*).



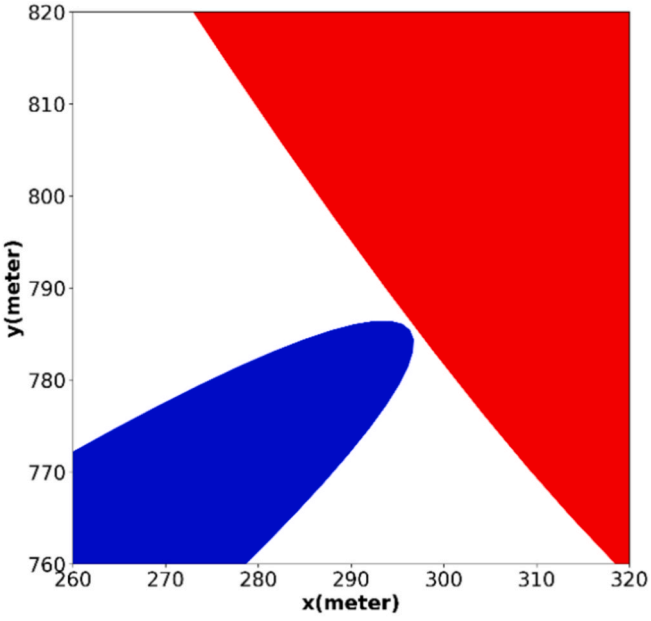
(i) Case3 of OS2



(j) Case4 of OS2

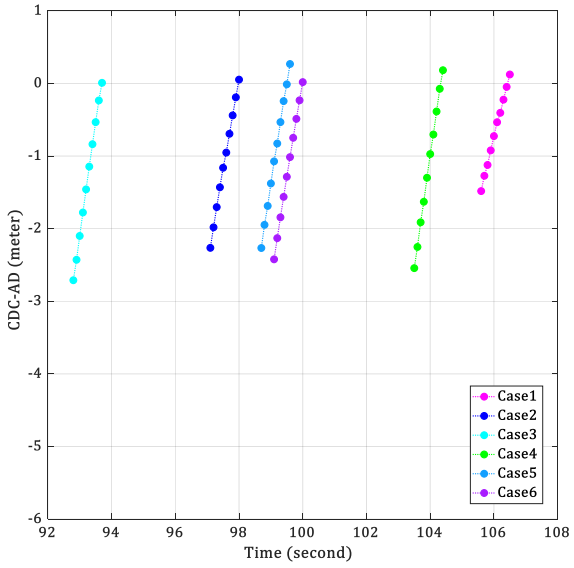


(k) Case5 of OS2

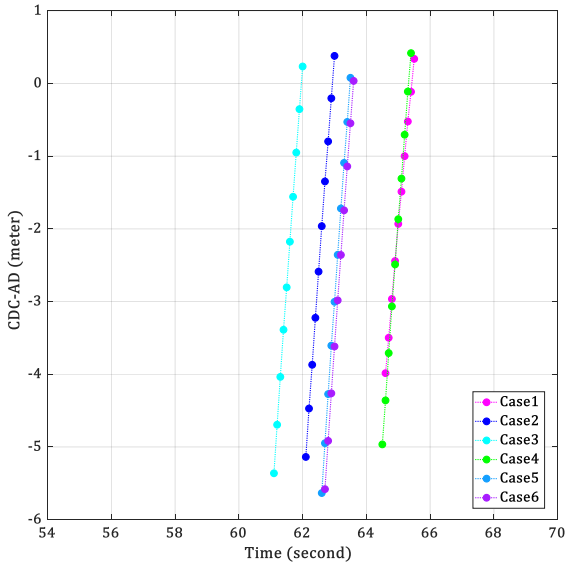


(l) Case6 of OS2

Fig. 15. (*continued*).



(a) Difference between CDC and AD values in OS1



(b) Difference between CDC and AD values in OS2

Fig. 16. Difference between CDC and AD

Furukawa (2022), who did not consider sea conditions, and our study, where sea conditions were considered, we concluded that θ_{ic} is about 28°–36° for this TSS.

3.4. Reliability of collision assessment

In this study, the timing of the ships' actions was determined by assessing whether or not the two ships will collide when approaching at close distances. In this case, verifying the accuracy of CDC is essential. Therefore, we verified whether the CA results provide the exact timing of collision evasive action by sampling, as shown in Fig. 14.

In Fig. 14, the black ellipse represents OS1, the blue ellipse represents OS2, and the red ellipse represents TS4. The past trajectories of the ships are shown. The horizontal and vertical axes represent the x- and y-coordinates, respectively, and the trajectories were plotted from the start of the simulation until 0.1s before t_c. Fig. 14 shows the situation between the two OSs and the largest (TS4) immediately before the collision. The two ships appear to collide, but the detailed situation cannot be observed because of the scale of the figure. The exact time of a collision in the presented situation must be known to verify the CDC and present the action timing of the ship. Therefore, Fig. 15 show the situation 0.1s before collision.

Fig. 15 shows only a portion of the ellipse because the figure is enlarged. The partial black, blue, and red ellipses represent OS1, OS2, and TS4, respectively. The horizontal and vertical axes are the x- and y-coordinates, respectively. The two ships have not yet contacted but appear to be colliding. As shown in panels (d) and (g) of Fig. 15, OS1 and OS2 do not make contact with TS4.

In the previous study (Lee and Furukawa, 2022), the model ship was only 2.5 m long. Therefore, whether the accuracy of CDC can be checked on CA of large ships is uncertain. Fig. 15 demonstrates that the CDC can be applied to the CA of large ships.

We further validated the CDC by comparing (rather than visualizing) the calculated values. Fig. 16 show the difference between the CDC and AD (d_{ca}) for each case at 0.1s intervals.

The horizontal axis is the time, and the vertical axis is d_{ca} in meters. d_{ca} was greater than 0 m when a collision occurred. Because OS1 had a longer t_s, it had a wider range of collision times than OS2.

The accuracy of CDC was further verified in situations of no collisions. Fig. 17 and Table 10 indicate that the CDC could be used to correctly determine when no collision occurred.

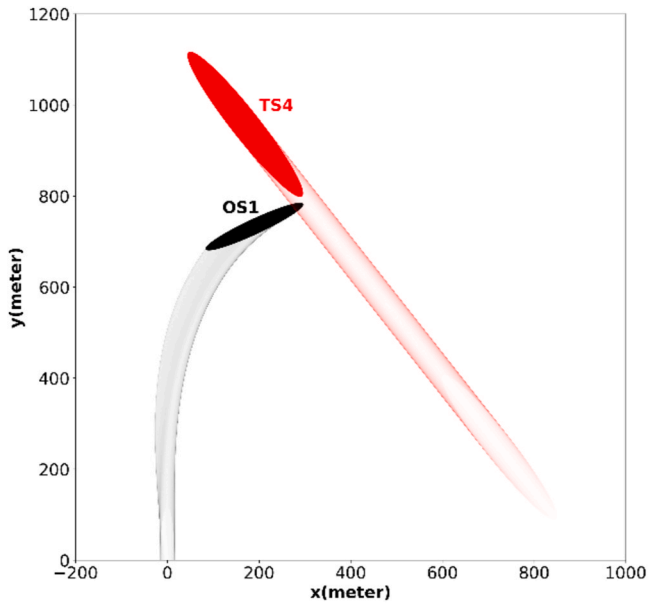
The horizontal and vertical axes represent the x- and y-coordinates, respectively. The black ellipse represents OS1, the blue ellipse represents OS2, and the red ellipse represents TS4. Fig. 17(a)–(c) show three non-collision cases. Case 2 for OS1 is shown in more detail in Fig. 17(c)–(h) because the two ships narrowly passed each other and appeared to collide at $t_s = 101.0$s. However, the closer look shows that the two ships narrowly passed without colliding. These results confirm that the CDC could be used to accurately determine non-collision cases, and the limitations pointed out by Lee and Furukawa (2022) could be corrected by adjusting the calculation interval time of CDC.

This analysis verifies the accuracy and reliability of CDC in several cases of no collision situations. Table 11 gives the CDC reliability results of the verification targeting the largest TS4.

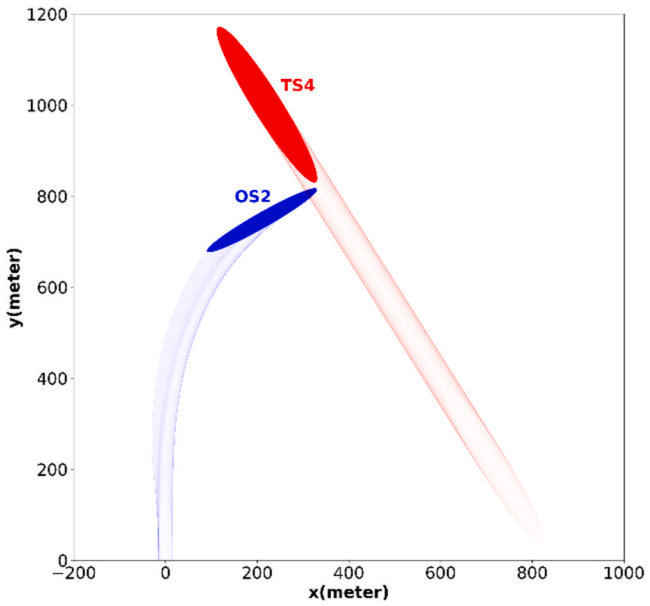
In Table 11, the CA in each case is presented at the closest AD at which a collision can occur between the OSs and TS4. OS1 did not collide with TS4 when d_{ca} was at least 12 m, and OS2 did not collide with TS4 when d_{ca} was at least 17 m.

3.5. Summary

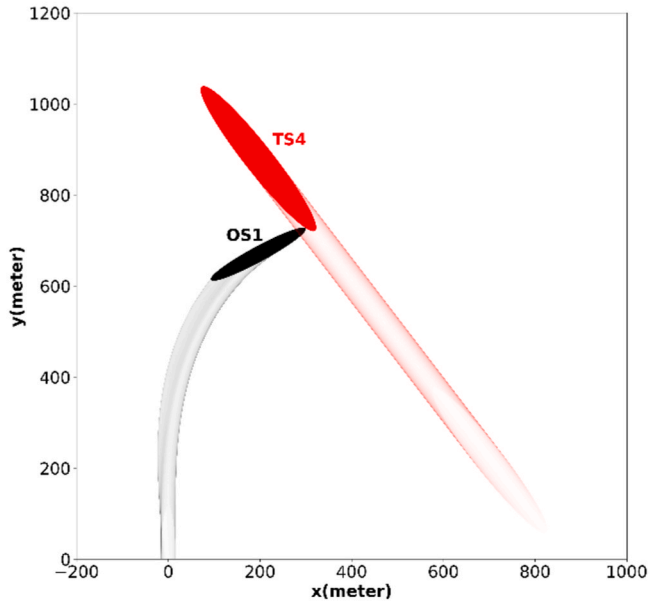
This study presents an engineering approach for quantifying ambiguous COLREGS. Here we summarize the main conclusions.



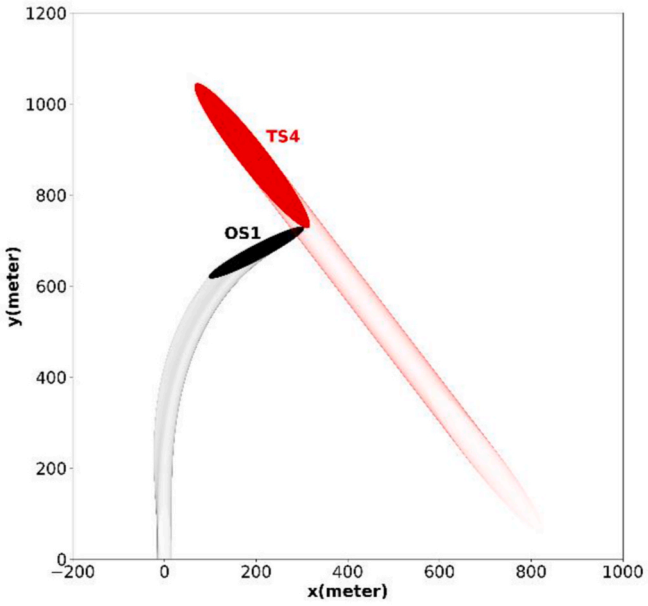
(a) Case1 of OS1 ($\theta_{ot} = 38°$)



(b) Case1of OS2 ($\theta_{ot} = 32°$)

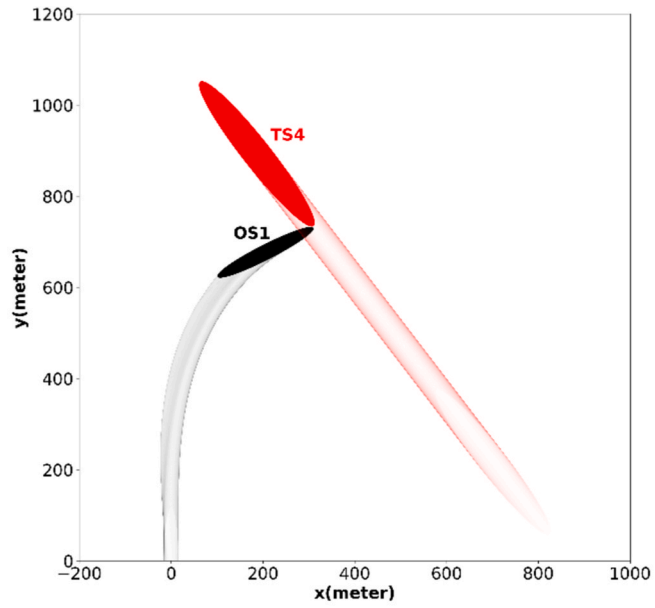


(c) Case2 of OS1($\theta_{ot} = 38°;\ t_s = 101.0$ s)

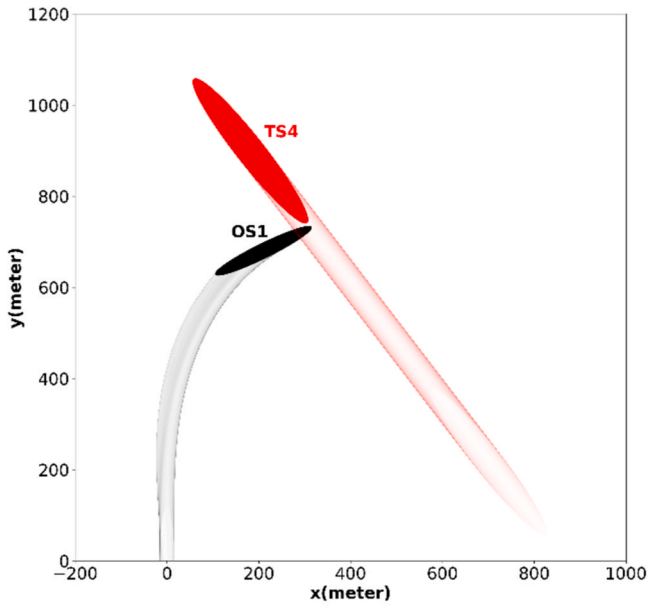


(d) Case2 of OS1 ($\theta_{ot} = 38°;\ t_s = 102.0$ s)

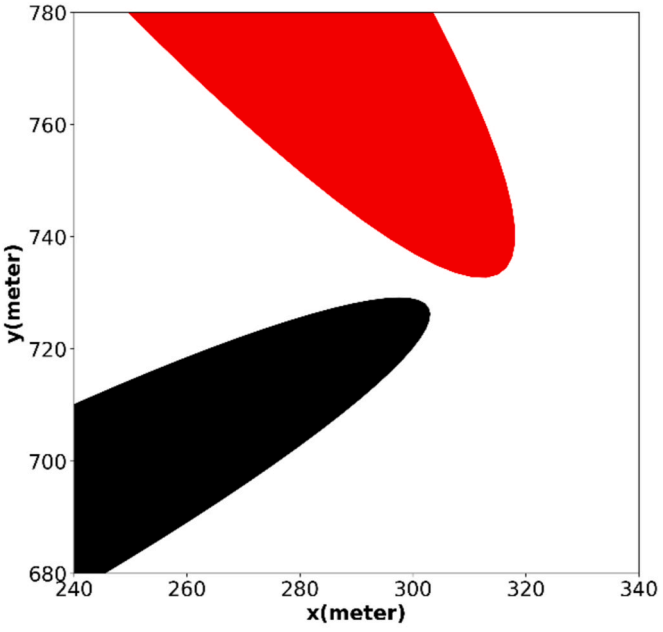
Fig. 17. Examples of a non-collision.



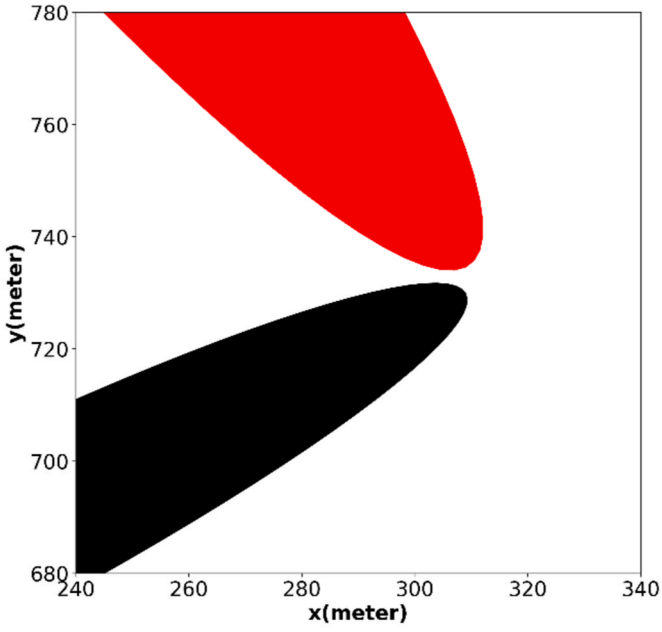
(e) Case2 of OS1($\theta_{ot} = 38°;\ t_s = 103.0$ s)



(f) Case2 of OS1 ($\theta_{ot} = 38°;\ t_s = 104.0$ s)



(g) Case2 of OS1 ($\theta_{ot} = 38°;\ t_s = 101.8$ s)



(h) Zoom-in on Case2 of OS1 ($t_s = 102.0$ s)

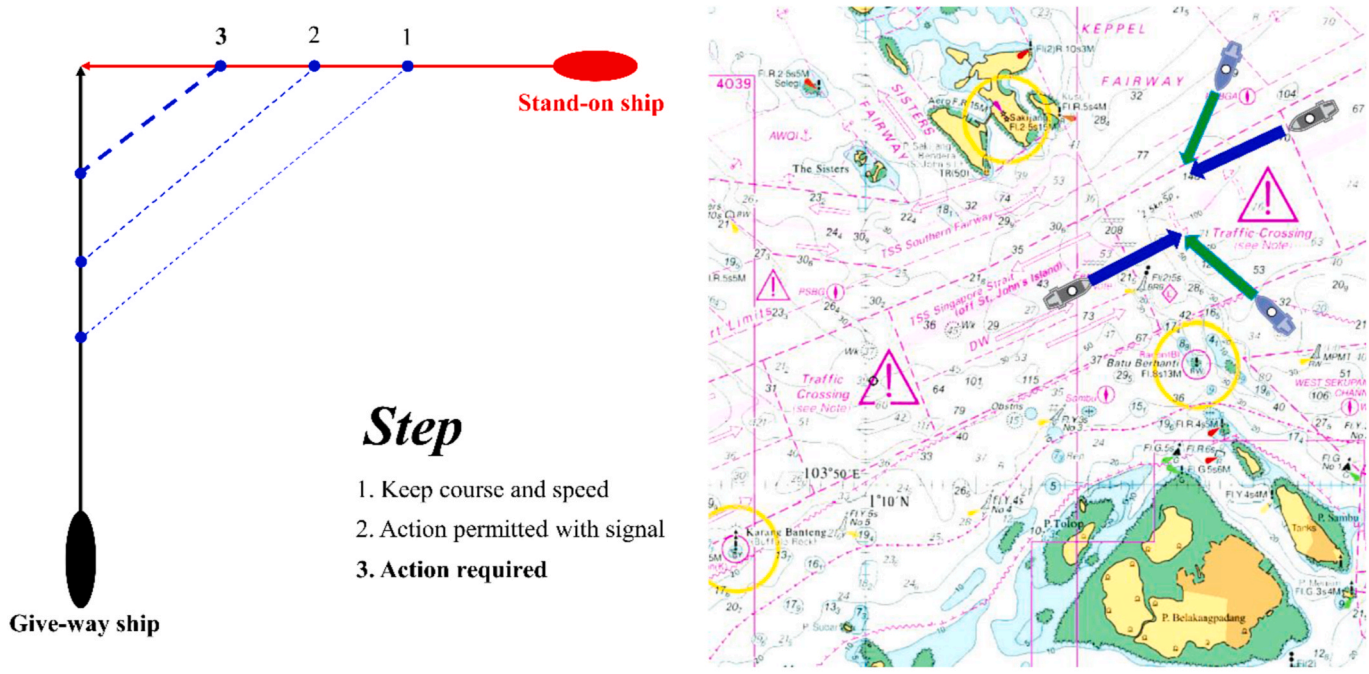
Fig. 17. (*continued*).

Table 10
Collision assessment results for Case 2 of OS1 ($\theta_{ot} = 38°$).

Time (s)	101.0	101.8	102.0	103.0	104.0
AD (m)	216.97	219.33	219.98	223.47	227.42
CDC (m)	212.16	213.97	Nan	Nan	Nan
d_{ca} (m)	−4.81	−5.36	Nan	Nan	Nan
CA result	No collision	No collision	No collision	No collision	No collision

Table 11
Collision assessment results.

	OS1 and TS4 (θ_{ot} = 38°)					
	Case1	Case2	Case3	Case4	Case5	Case6
Time(s)	91.5	95.6	92.4	104.2	98.2	98.1
AD (m)	211.84	209.39	204.87	204.81	209.21	211.89
CDC (m)	189.47	189.64	188.16	192.52	192.44	192.58
d_{ca} (m)	−22.37	−19.75	−16.71	−12.29	−16.77	−19.31
CA result	No collision	No collision	No collision	No collision	No collision	No collision
	OS2 and TS4					
	Case1 (θ_{ot} = 30°)	**Case2 (θ_{ot} = 32°)**	**Case3 (θ_{ot} = 30°)**	**Case4 (θ_{ot} = 30°)**	**Case5 (θ_{ot} = 30°)**	**Case6 (θ_{ot} = 30°)**
Time(s)	62.0	62.8	61.9	65.1	63.2	63.4
AD (m)	245.60	235.63	239.18	242.01	241.62	241.82
CDC (m)	224.06	217.74	222.00	222.93	221.25	221.85
d_{ca} (m)	−21.54	−17.89	−17.18	−19.08	−20.37	−19.97
CA result	No collision	No collision	No collision	No collision	No collision	No collision



(a) Suggested actions for a step-by-step stand-on ship action (Cockcroft and Lameijer, 2003)

(b) Singapore Main Strait

Fig. 18. Summary of result.

Table 12
d_i between the two ships at $max\ \theta_{ic}$.

		d_i			
		TS1	TS2	TS3	TS4
Case 1	**OS1**	367 m	528 m	608 m	726 m
	OS2	377 m	517 m	609 m	701 m
Case 2	**OS1**	395 m	550 m	627 m	703 m
	OS2	419 m	557 m	602 m	692 m
Case 3	**OS1**	388 m	541 m	616 m	690 m
	OS2	378 m	518 m	611 m	657 m
Case 4	**OS1**	423 m	590 m	672 m	754 m
	OS2	393 m	539 m	587 m	683 m
Case 5	**OS1**	408 m	569 m	648 m	726 m
	OS2	386 m	519 m	576 m	671 m
Case 6	**OS1**	409 m	570 m	650 m	728 m
	OS2	386 m	529 m	577 m	671 m

Table 13
Situations of inevitable collision between two ships and t_{action}.

		Types of ships			
	Parameter	TS1	TS2	TS3	TS4
OS1	θ_{ic}	20 °	28 °	32 °	36 °
	τ_{action}	73.4–82.3 s	83.9–94.2 s	89.2–99.7 s	93.7–104.4 s
	d_i	367–423 m	528–590 m	608–672 m	703–754 m
OS2	θ_{ic}	16 °–18 °	22 °–24 °	24 °–26 °	28 °–30 °
	τ_{action}	49.3–51.9 s	56.0–59.3s	58.4–60.9 s	61.8–65.3 s
	d_i	377–419 m	517–557 m	576–611 m	671–701 m

Interpreting COLREGS rules 8, 16, and 17 together, when two ships are in close situation, the stand-on ship should take cooperative action to avoid collision and the give-way ship should do everything possible to avoid collision. This study applied CDC to CA of two nearby ships, from which we can infer the collision avoidance point of the ship.

In this regard, a proposal for the action of the stand-on ship when two ships are in close situation is shown in Fig. 18.

As shown in Fig. 18 (a), the action steps of the stand-on ship are divided but no specific distances are given in each step (Cockcroft and Lameijer, 2003). In this study, we wanted to provide a numerical value for the third step, which requires the action of the stand-on ship, and to know when the give-way ship must use engine control to avoid collision. Therefore, we identified the inevitable collision between OS and TS that may occur as shown in Fig. 18(b) in the ***Singapore Main Strait*** and proposed when the stand-on ship should perform cooperative action and when the give-way ship should use its engine to avoid the collision. Based on the scenario and the CDC analysis, Table 12 presents the timings for the actions of the give-way ship (OS) and stand-on ship (TS), which can be determined from d_i at $max\ \theta_{ic}$.

d_i varied with the size of the TS. Among the different cases, the maximum d_i was 754 m for OS1 and 701 m for OS2. Although θ_{ic} did not depend on the sea conditions, d_i differed by up to 70 m under different sea conditions and Δd_i widely varied by TS length. In Case 4, the maximum d_i was 423 m for OS1, but 754 m for TS4. It can be concluded that larger TS encounters require earlier action by the OS.

Table 13 summarizes the results of collision avoidance according to the type of OS and TS. The results of this study are difficult to generalize, but if θ_{ic} and d_i are similar under different sea conditions excluding bad weather, the timing for actions (t_{action}) by the give-way ship and stand-on ship should be valid for the ***Singapore Main Strait***.

According to the values in Table 13, OS1 and OS2 must perform cooperative collision avoidance actions for the stand-on ship and speed control for the give-way ship. More specifically, for a given d_i and θ_{ic} OS1 and OS2 must take actions between 73 and 104 s and between 49 and 65 s, respectively.

4. Discussion and conclusions

Herein, we presented the timing for the evasive actions that must be taken by give-way and stand-on ships to avoid collision in the ***Singapore Main Strait***. A bulk carrier and container ship were considered as the OS, while ships passing through the ***Singapore Main Strait*** were used to set the length and width of different TSs. The CDC was used for CA between ships, which yielded θ_{ic} values. We expected θ_{ic} to vary depending on the sea conditions; however, no considerable difference was observed, which we attributed to the short $t_{\psi=90^\circ}$. The maximum θ_{ic} values appeared to be between 28° and 36° based on the ship types and sea conditions in the ***Singapore Main Strait***.

However, we did not consider the geographic characteristics of the TSS, such as the channel width and shallow water areas. If the water around the TSS is shallow, the OS may not be able to change its course much to avoid the TS, which limits the available rudder angle. Future research should consider the realistic degree to which the OS can change its rudder angle while navigating through the TSS of the ***Singapore Main Strait***.

The simulated ship trajectories were calculated based on a program developed by KRISO and STR. However, this program does not consider interactions between the OS and TS. The proximity of two ships can affect their trajectories, but this interaction was not considered in our simulation. In addition, our study was based on simulated ship trajectories, which may introduce some uncertainty.

However, because of the short simulation time, we do not anticipate significant differences between the simulated and actual ship trajectories. It is important to recognize these limitations and further investigate the ship-to-ship interactions and actual ship trajectories to improve the accuracy of future studies. This study and the previous study (Lee and Furukawa, 2022) focused on bulk carriers, VLCCs, and container carriers. Container carriers normally have the good turning performance, but bulk carriers and VLCCs have poor turning performances (Lee et al., 2021). These results suggest that a range of θ_{ic} between ships of different types can be partially determined. However, further research is required to account for ship-to-ship interactions and uncertainty in the ship trajectories. In addition, more ship types should be considered to calculate the θ_{ic} range in more detail. In addition, we need to study the required safety margin for actual MASSs to pass through the TSS.

Funding

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. 2022R1A6A3A01087185126178206434010 2).

CRediT authorship contribution statement

Hee-Jin Lee: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft. **Deuk-Jin Park:** Conceptualization, Data curation, Formal analysis, Methodology, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

Acknowledgment

[Inserts statement].

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